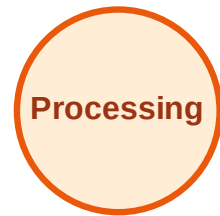


BFS Traversal

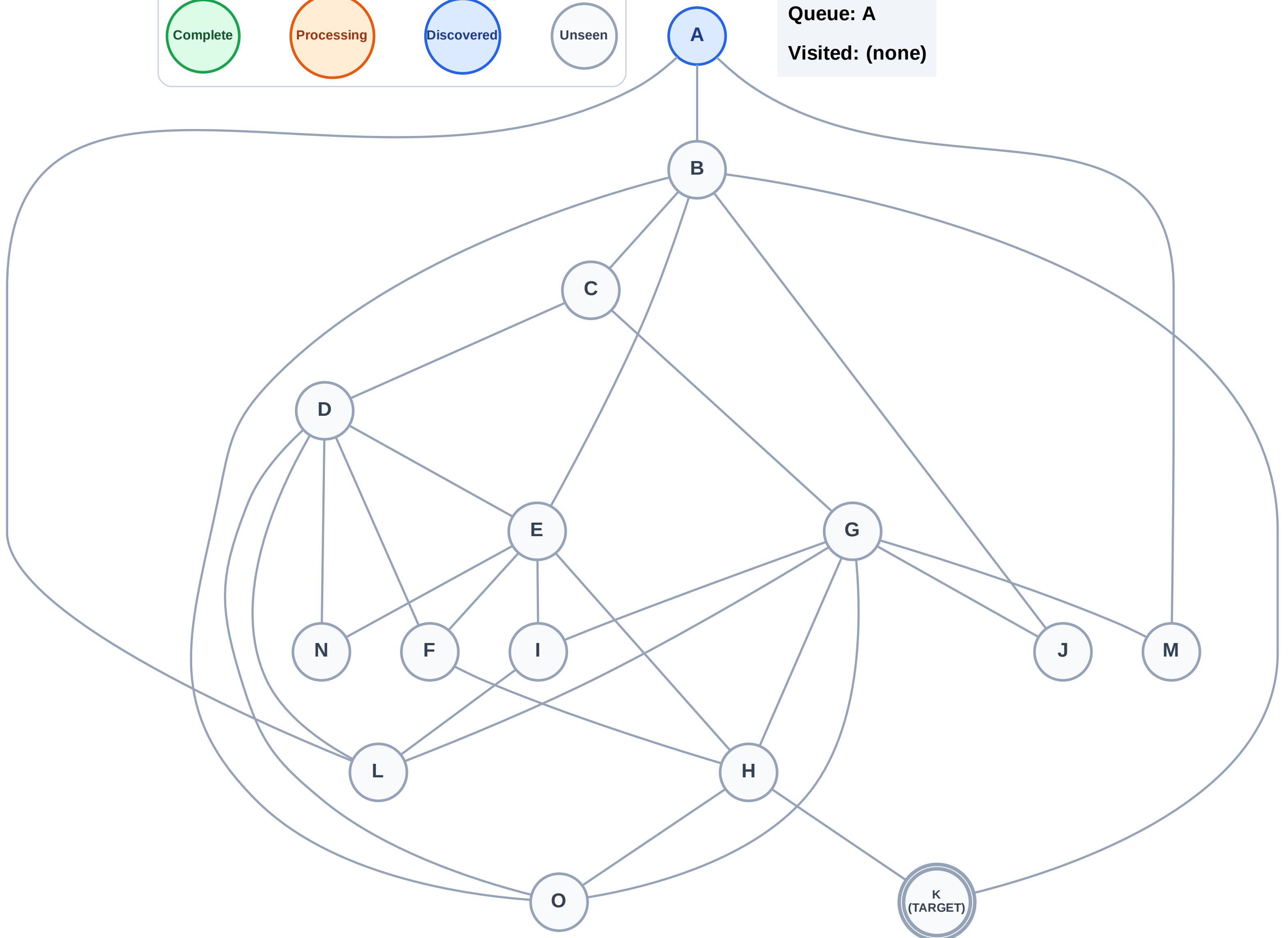
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

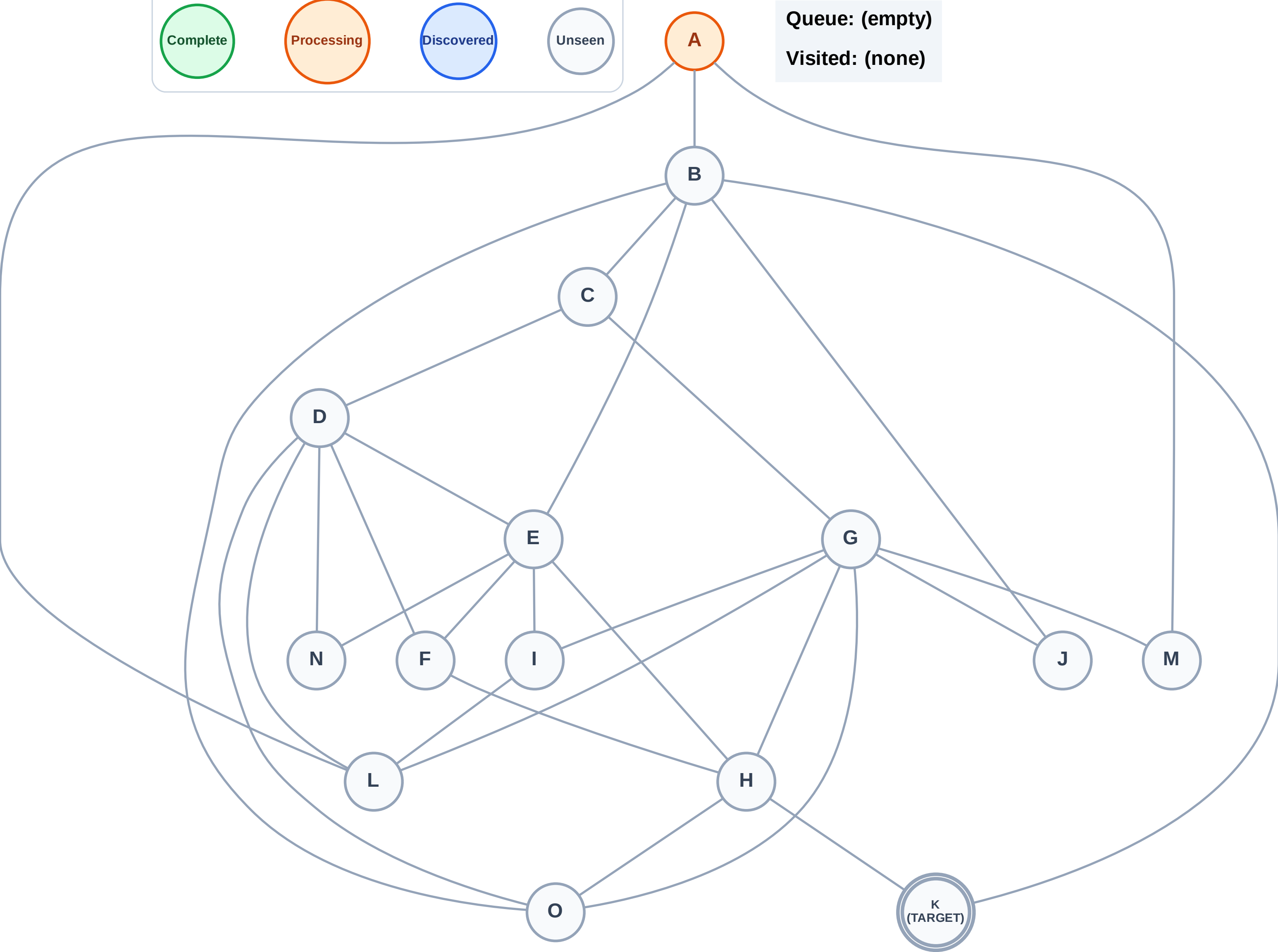
Complete

Processing

Discovered

Unseen

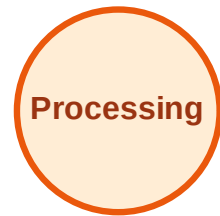
Queue: (empty)
Visited: (none)



BFS Traversal

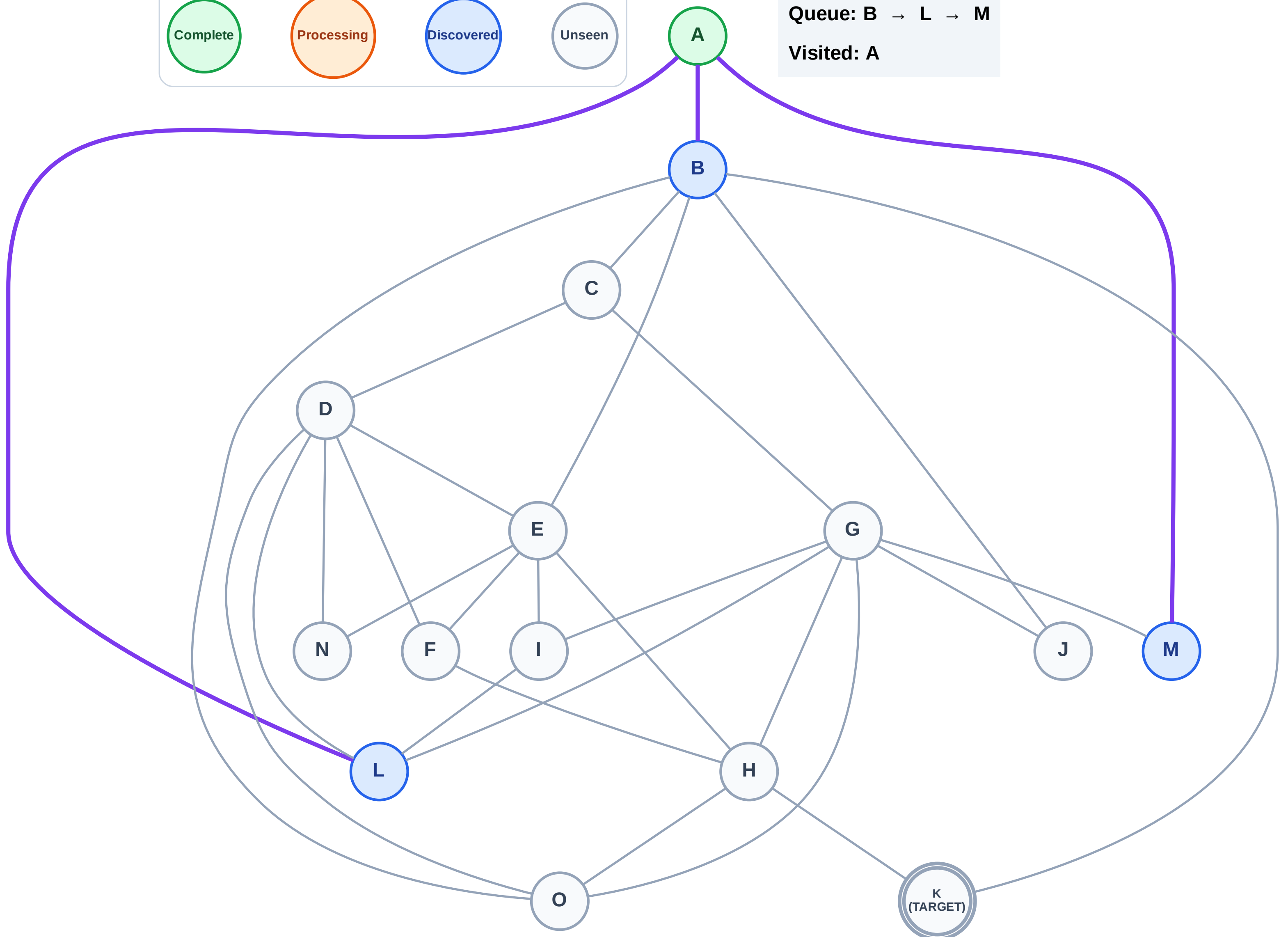
Step 3: Discover B, L, M from A

Legend



Queue: B → L → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

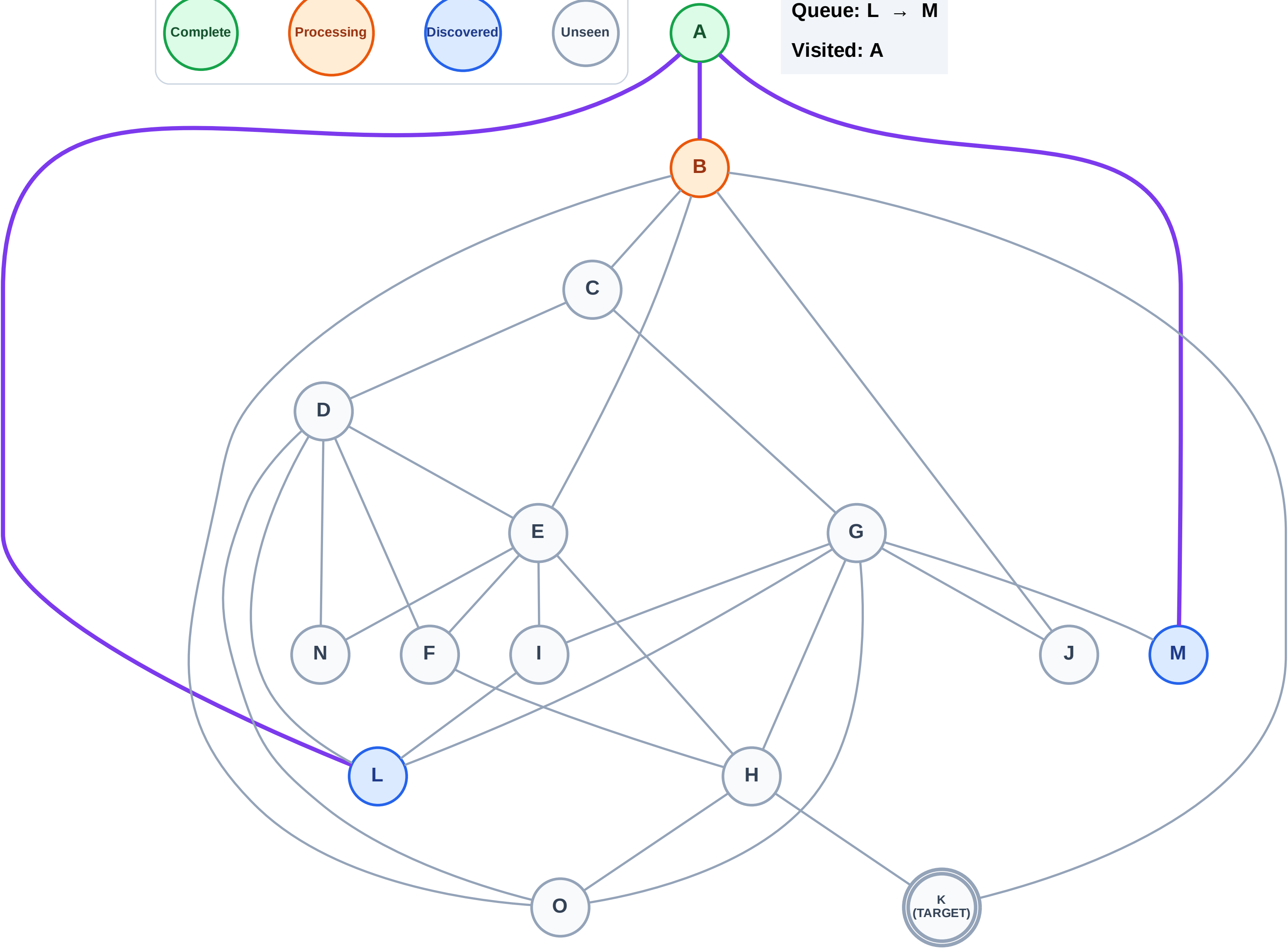
Processing

Discovered

Unseen

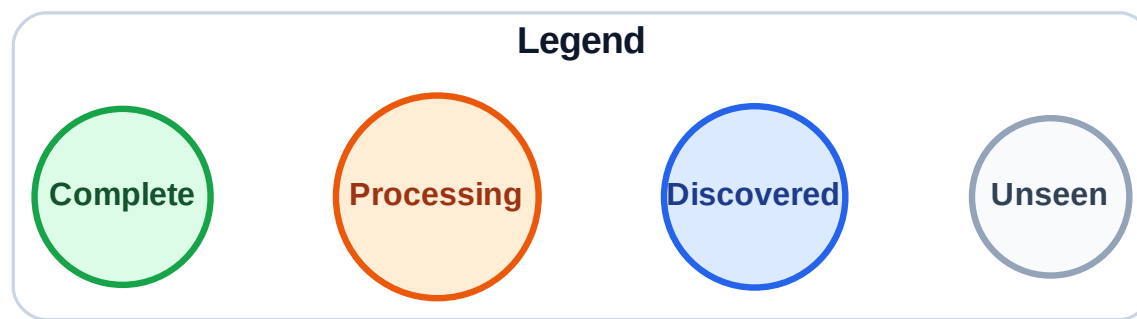
Queue: L → M

Visited: A



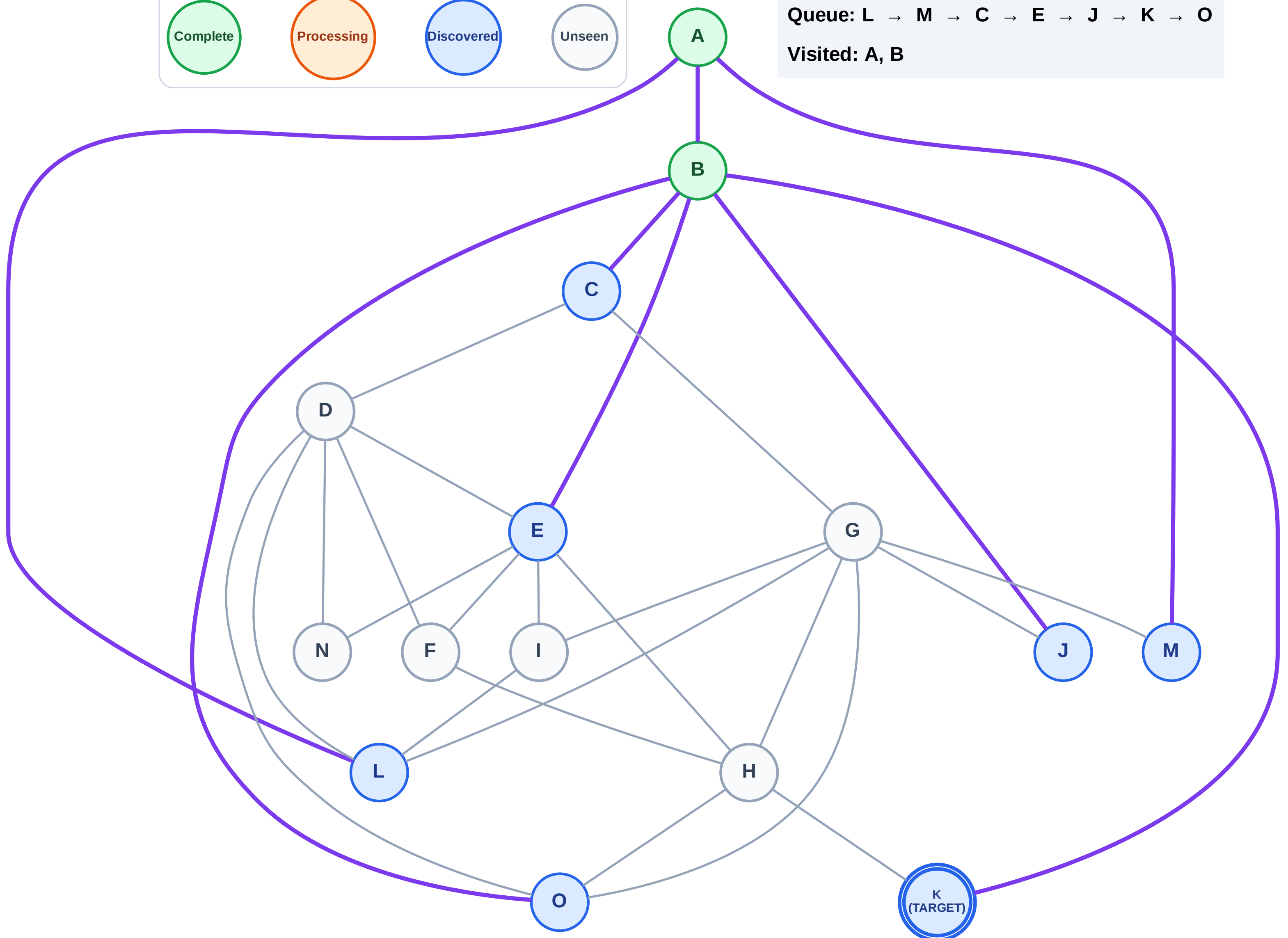
BFS Traversal

Step 5: Discover C, E, J, K, O from B



Queue: L → M → C → E → J → K → O

Visited: A, B



BFS Traversal

Step 6: Process node L

Legend

Complete

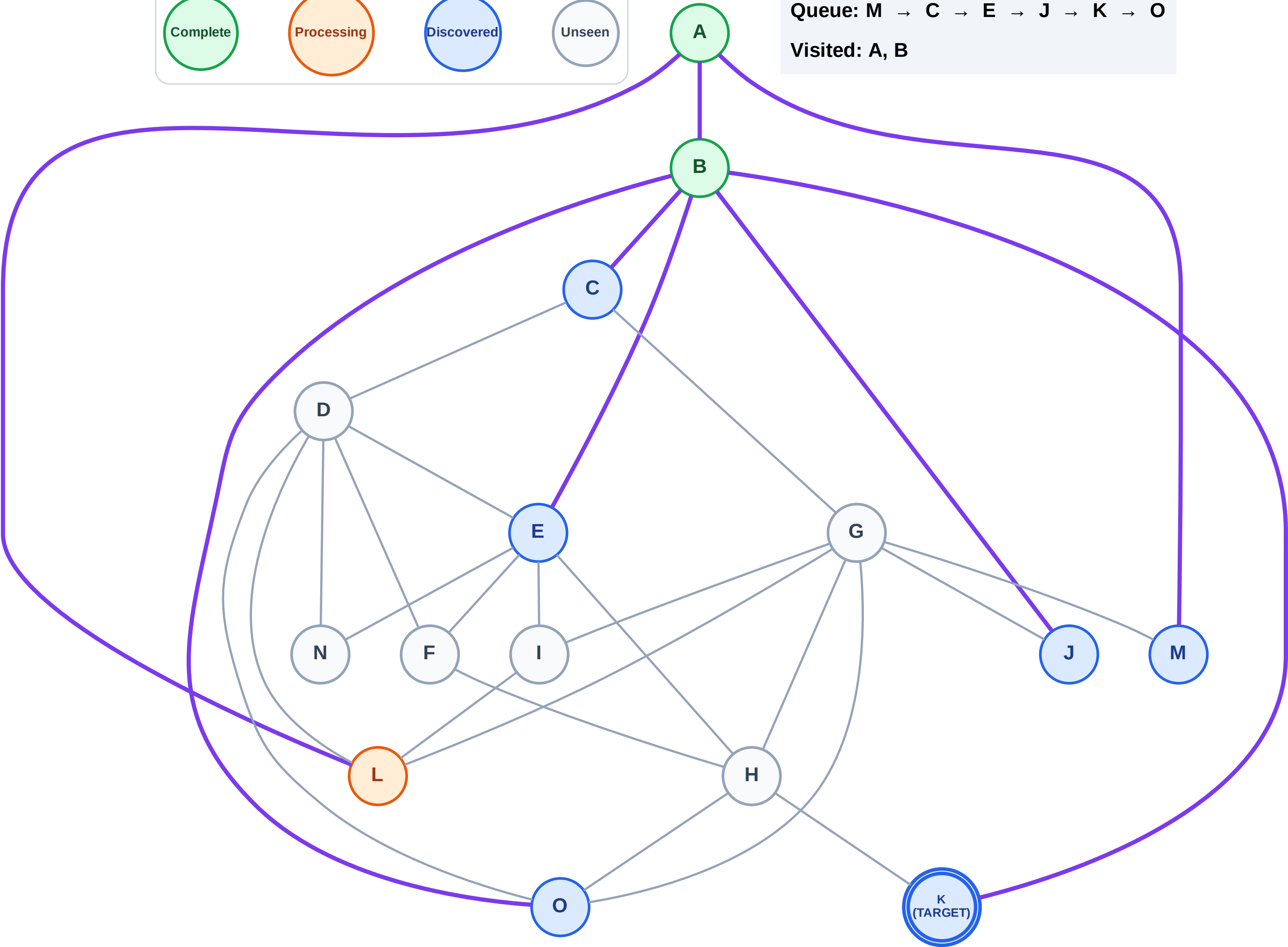
Processing

Discovered

Unseen

Queue: M → C → E → J → K → O

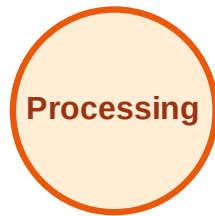
Visited: A, B



BFS Traversal

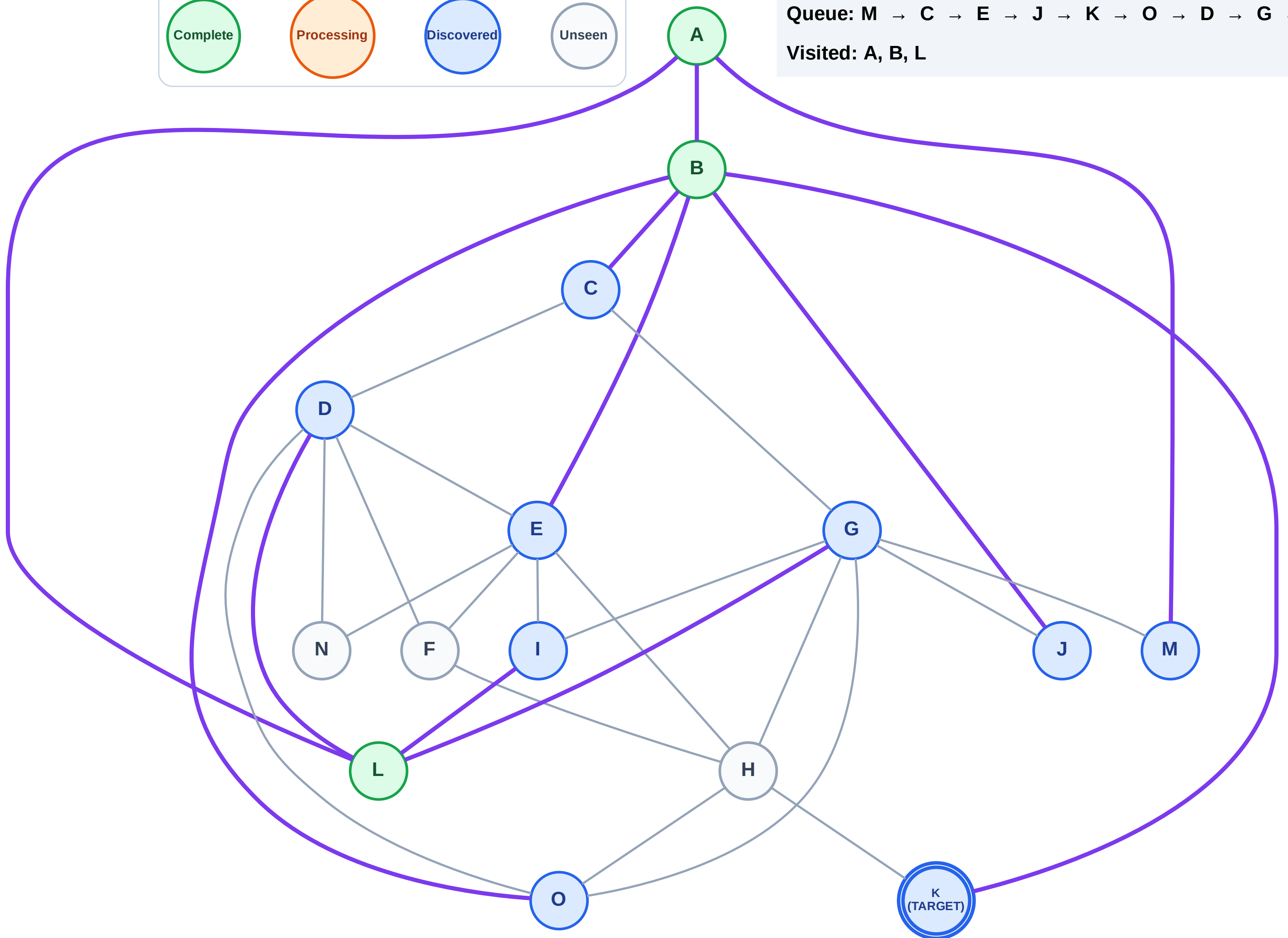
Step 7: Discover D, G, I from L

Legend



Queue: M → C → E → J → K → O → D → G → I

Visited: A, B, L



BFS Traversal

Step 8: Process node M

Legend

Complete

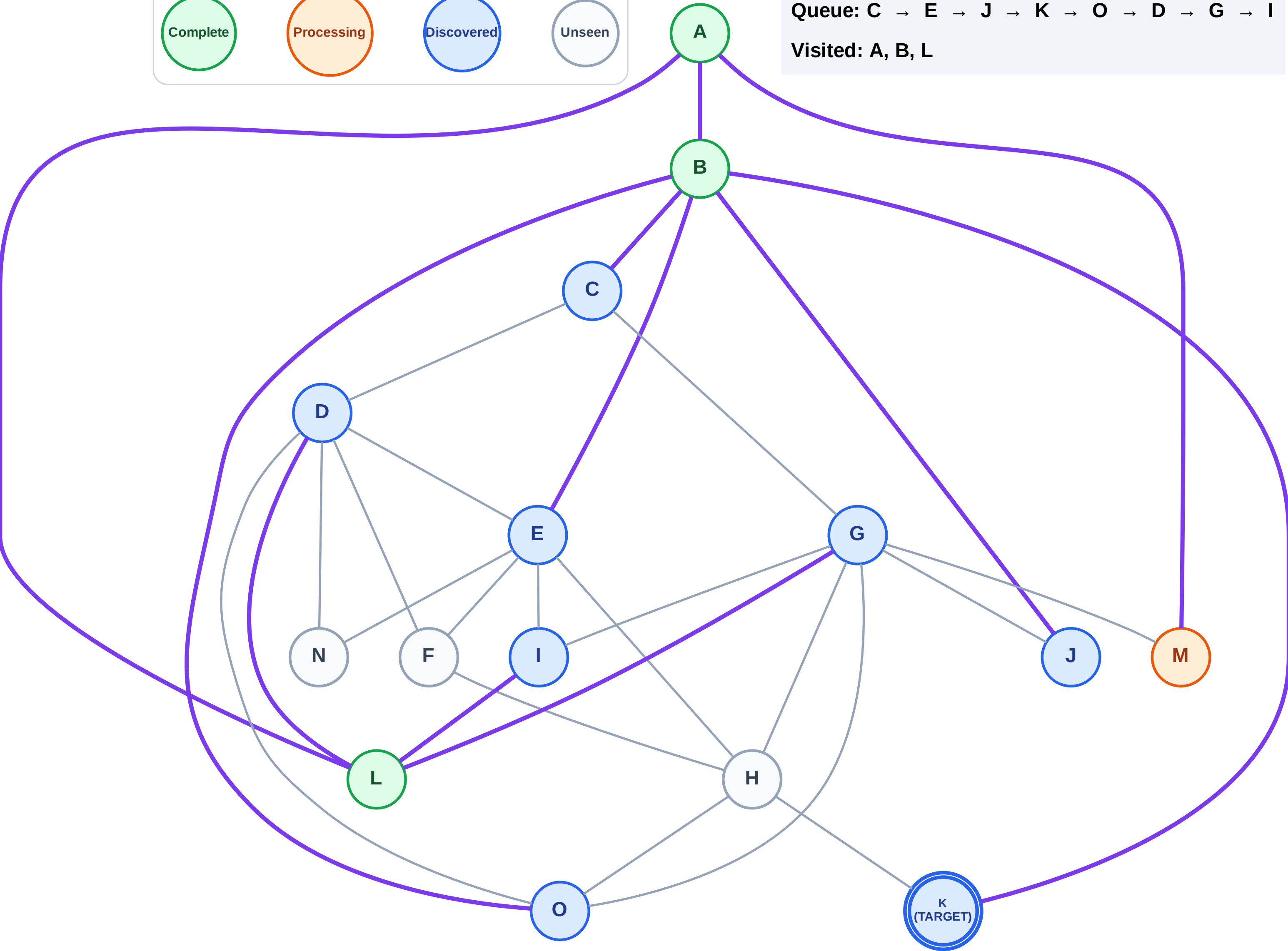
Processing

Discovered

Unseen

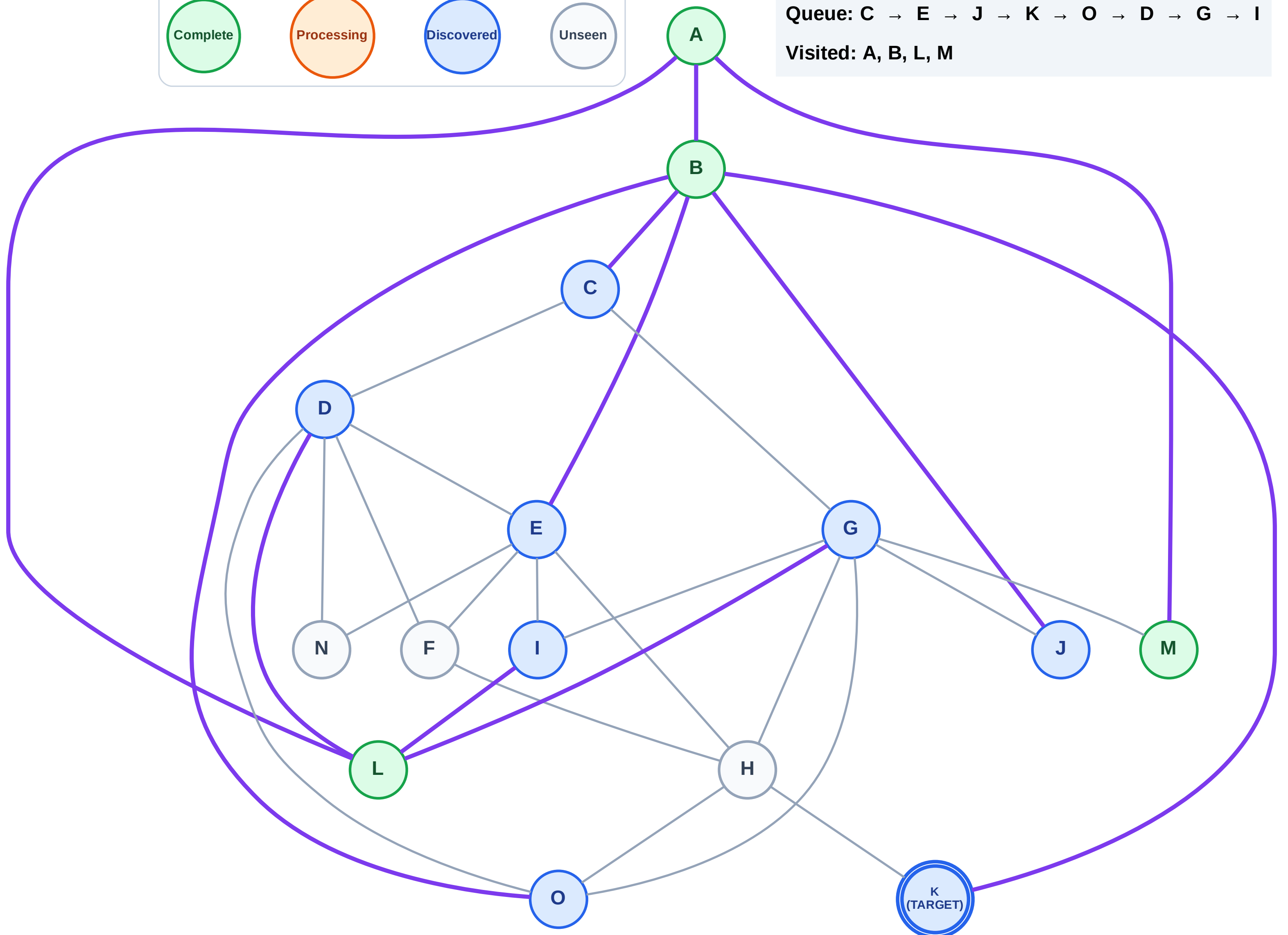
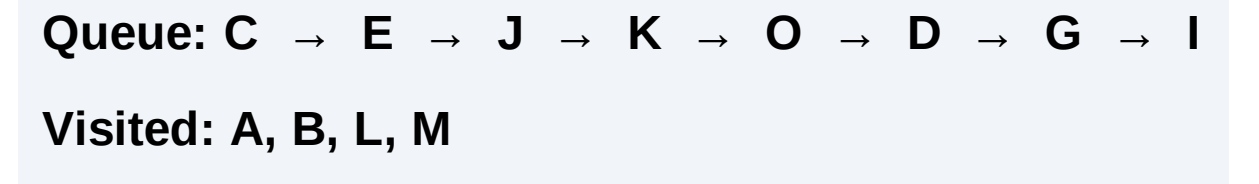
Queue: C → E → J → K → O → D → G → I

Visited: A, B, L



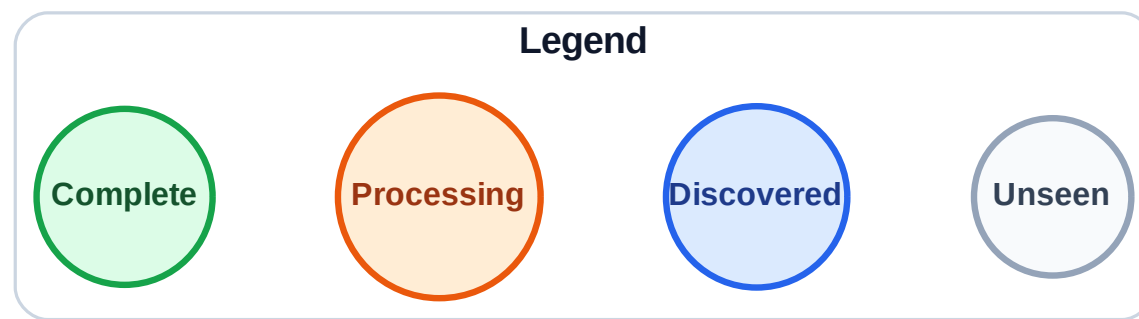
Step 9: No new neighbors from M

Step 9: No new neighbors from M



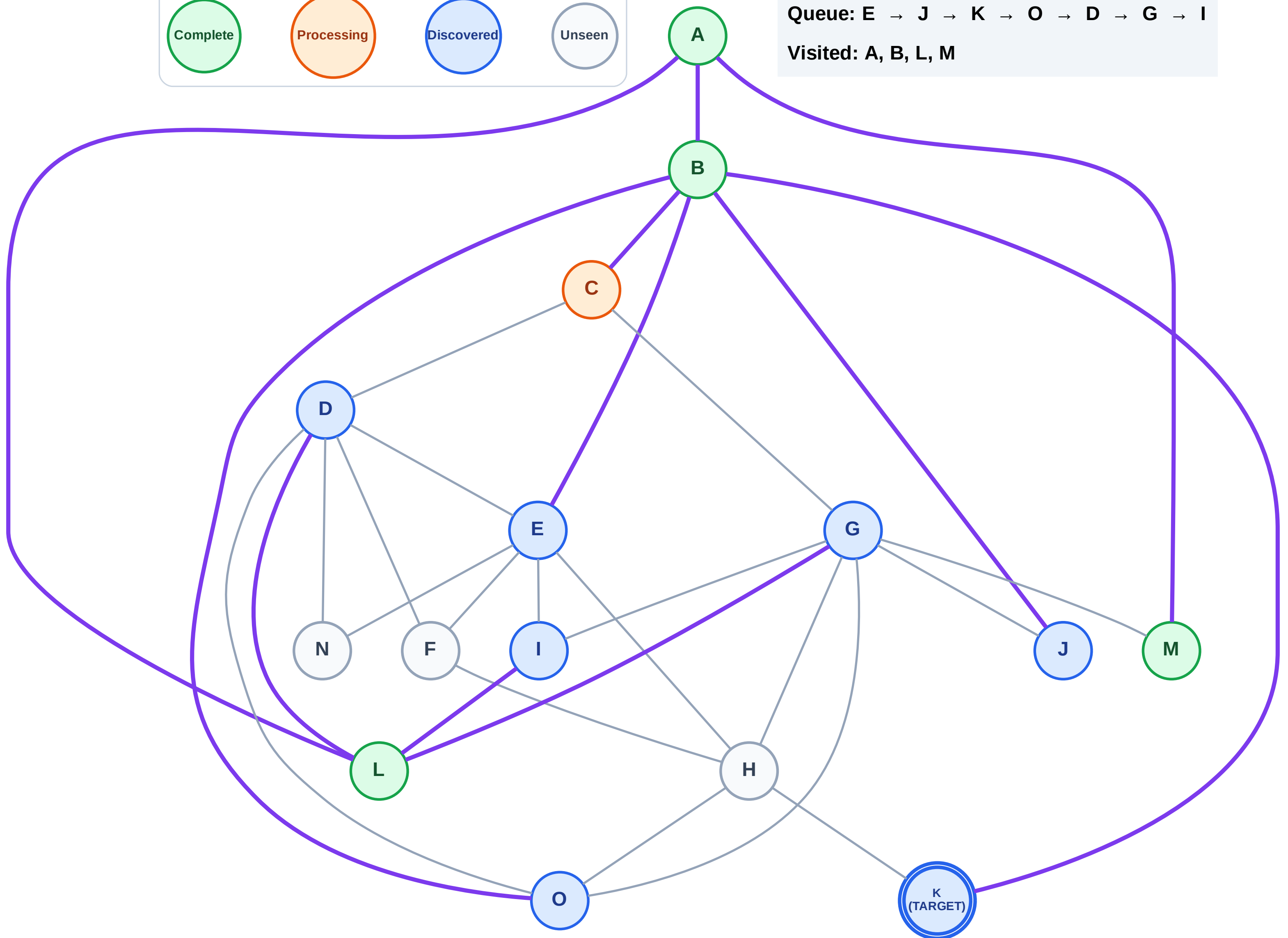
BFS Traversal

Step 10: Process node C



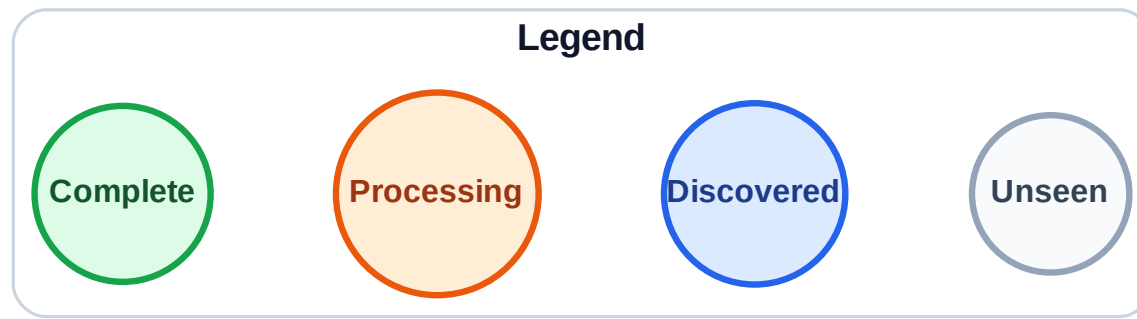
Queue: E → J → K → O → D → G → I

Visited: A, B, L, M



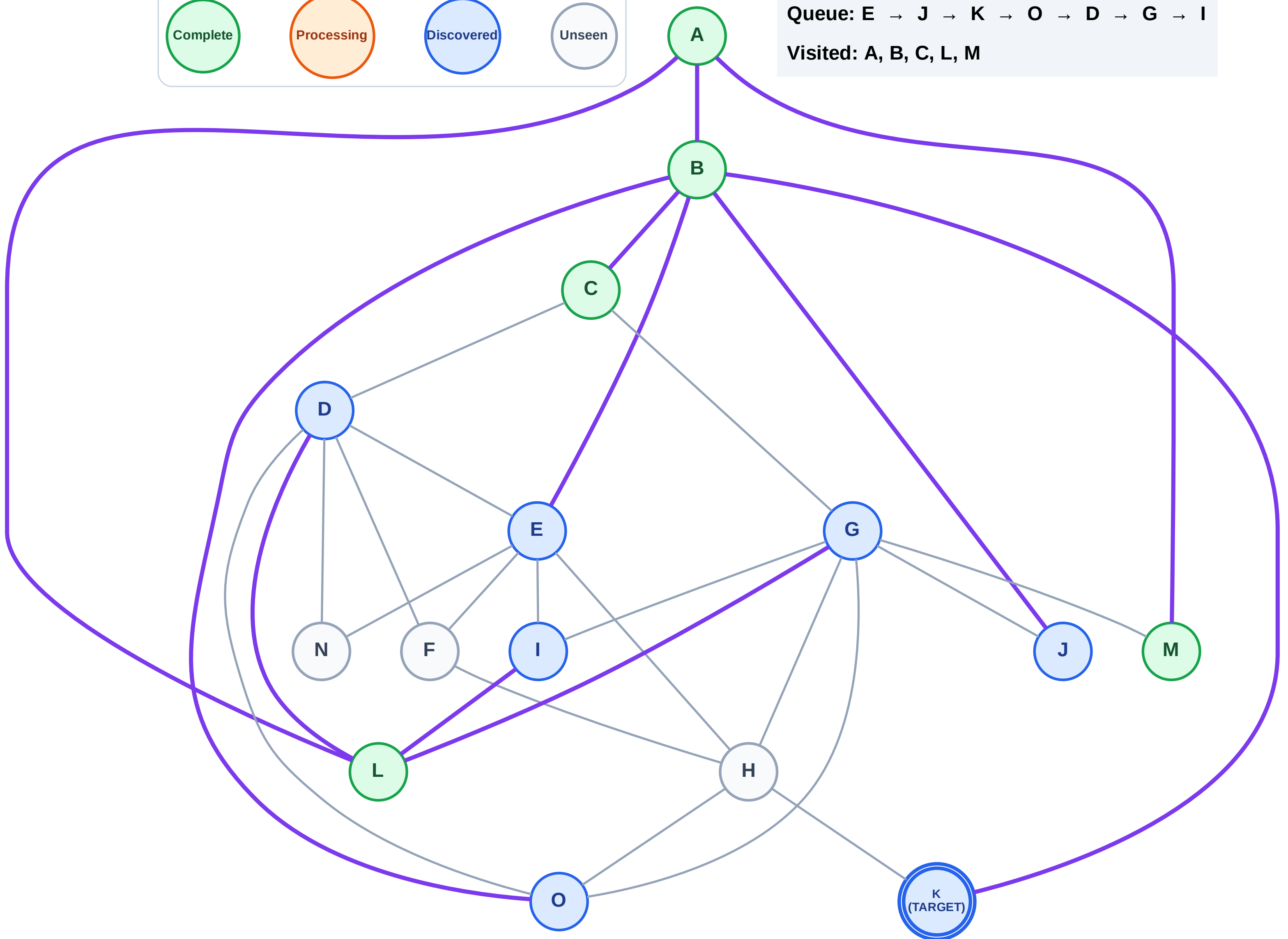
BFS Traversal

Step 11: No new neighbors from C



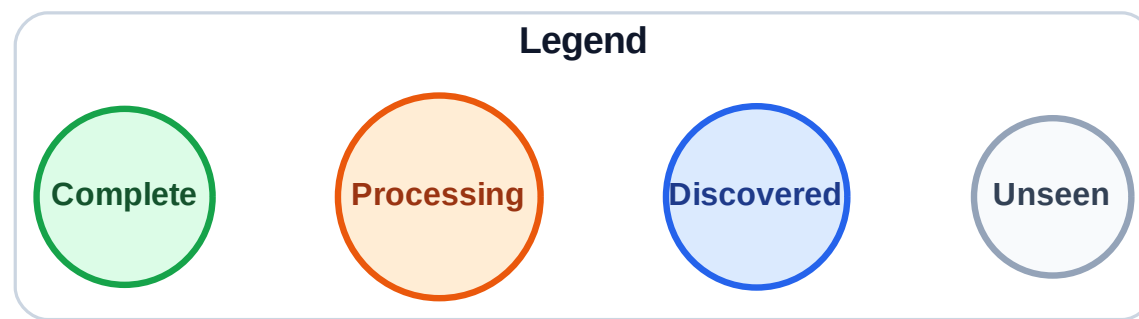
Queue: E → J → K → O → D → G → I

Visited: A, B, C, L, M

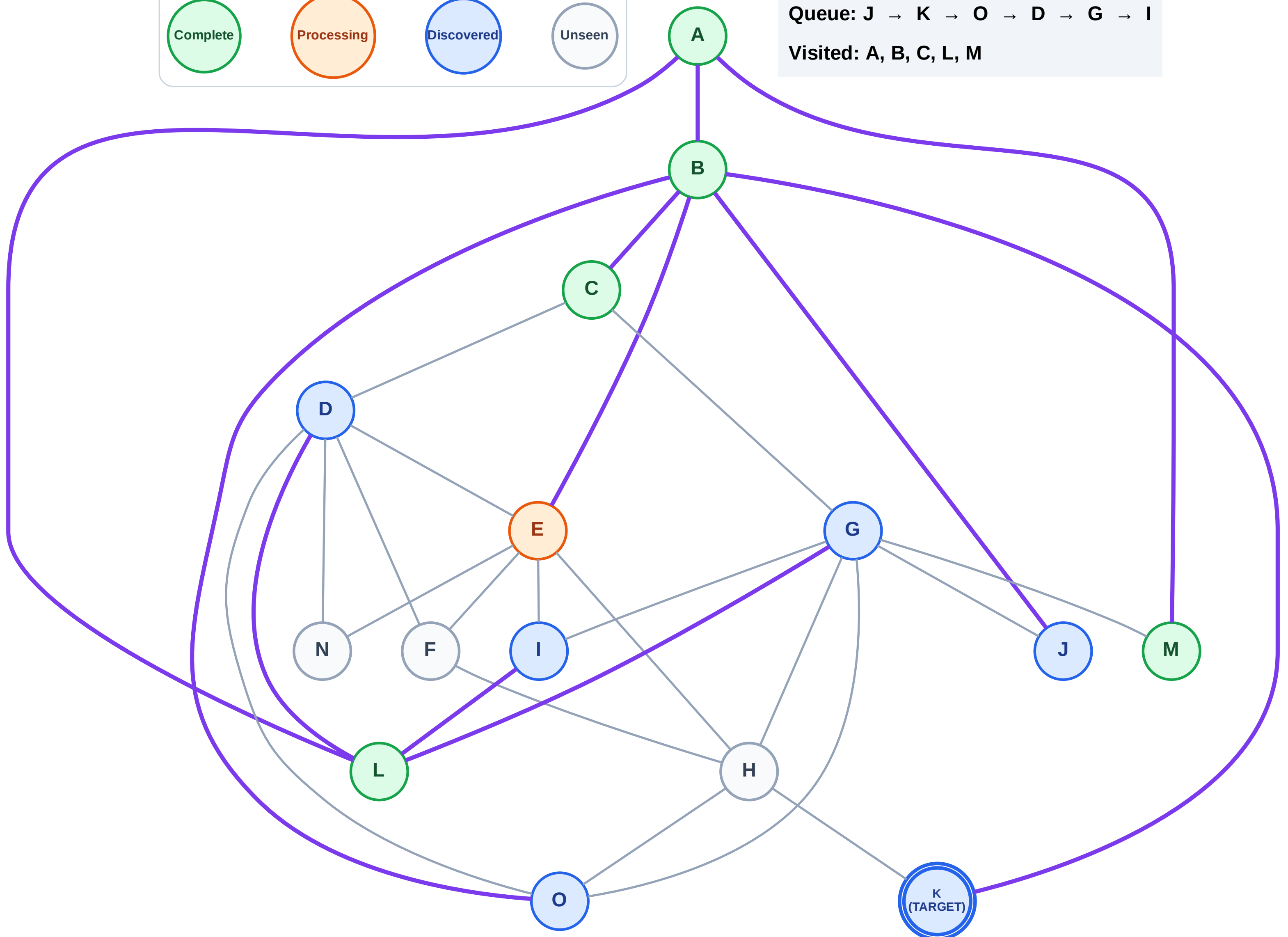


BFS Traversal

Step 12: Process node E

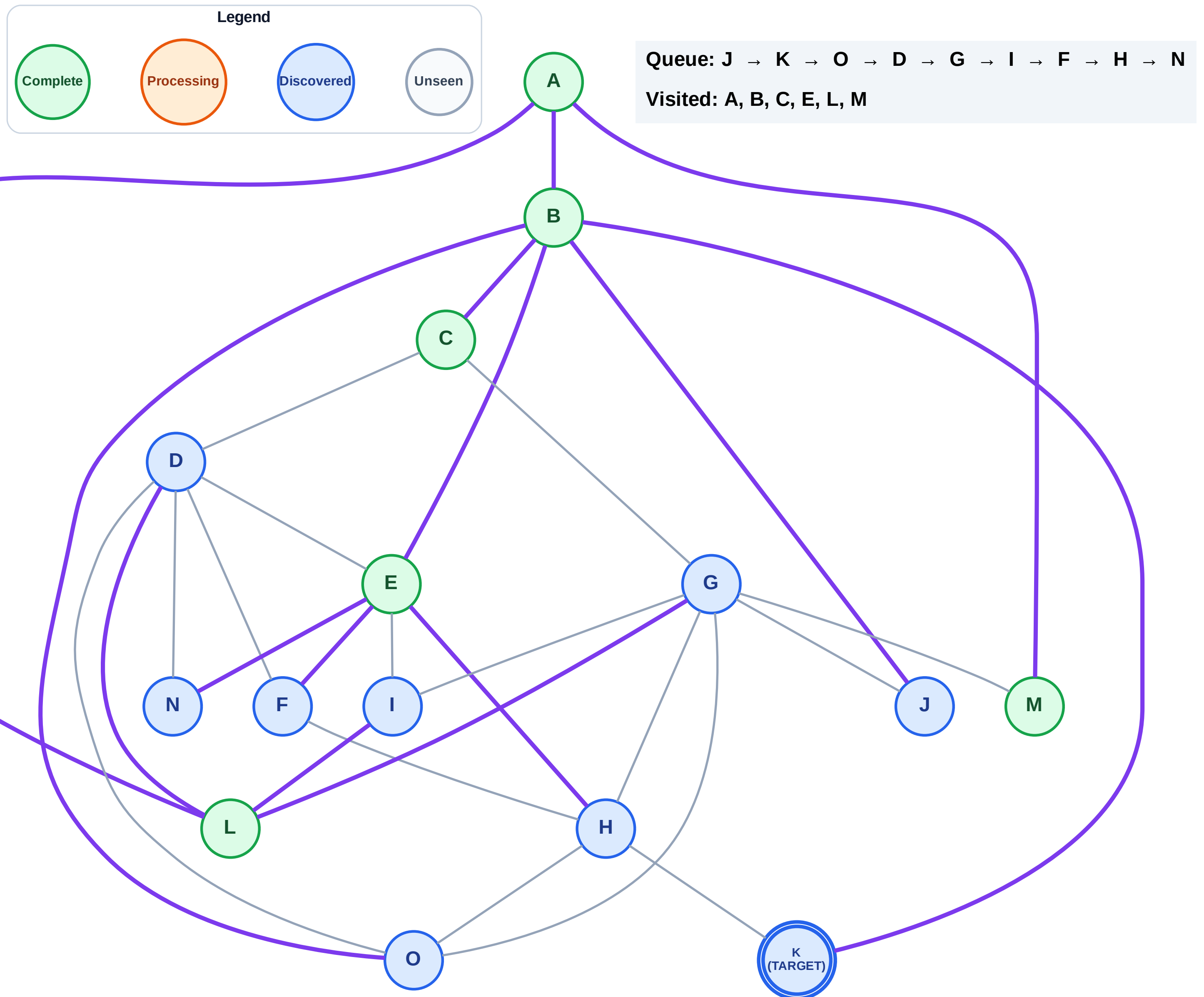


Queue: J → K → O → D → G → I
Visited: A, B, C, L, M



Step 13: Discover F, H, N from E

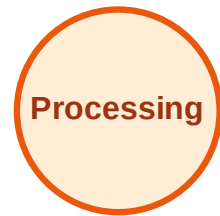
Step 13: Discover F, H, N from E



BFS Traversal

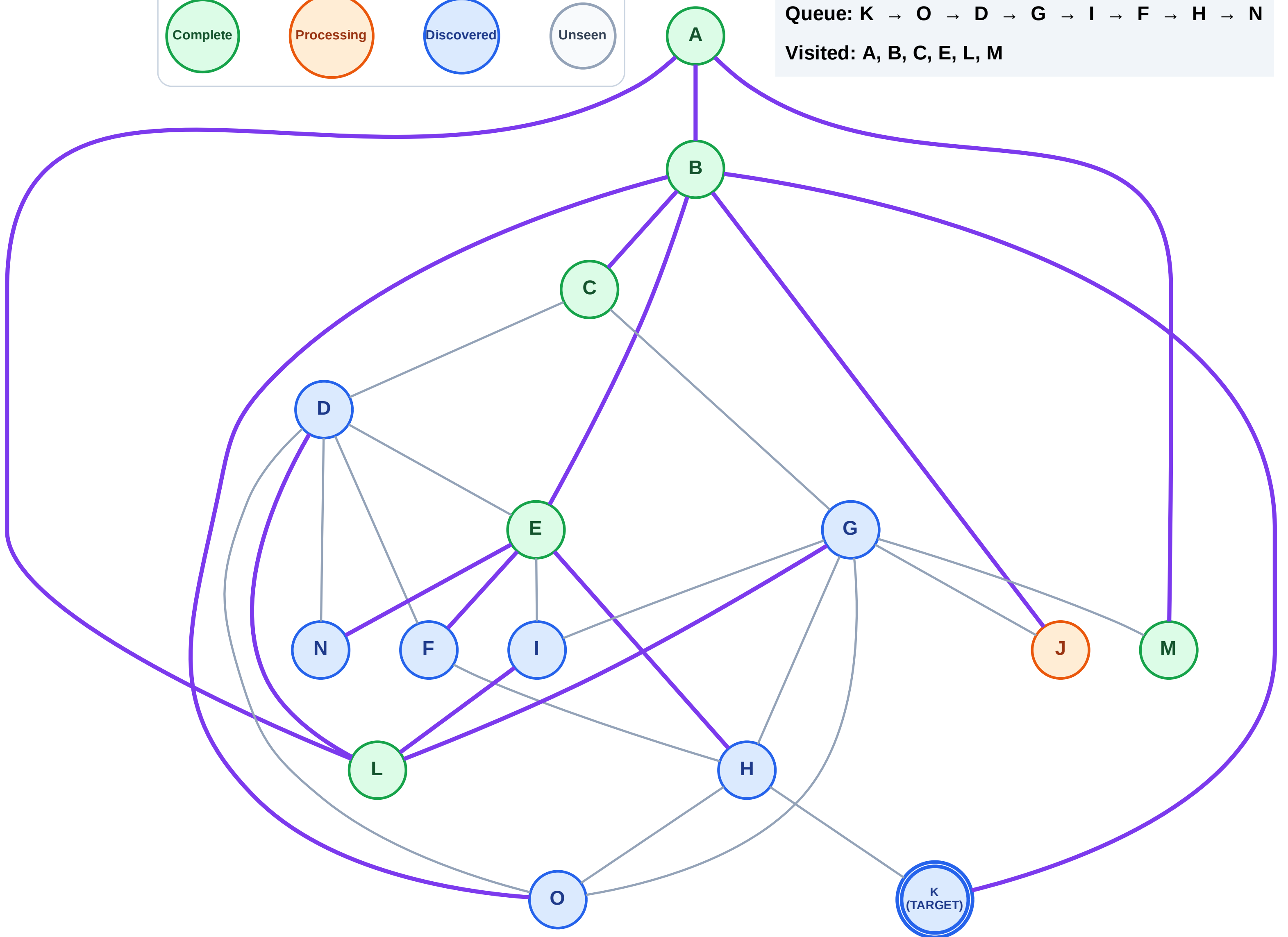
Step 14: Process node J

Legend



Queue: K → O → D → G → I → F → H → N

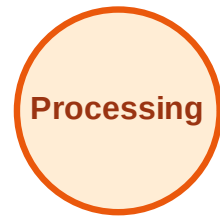
Visited: A, B, C, E, L, M



BFS Traversal

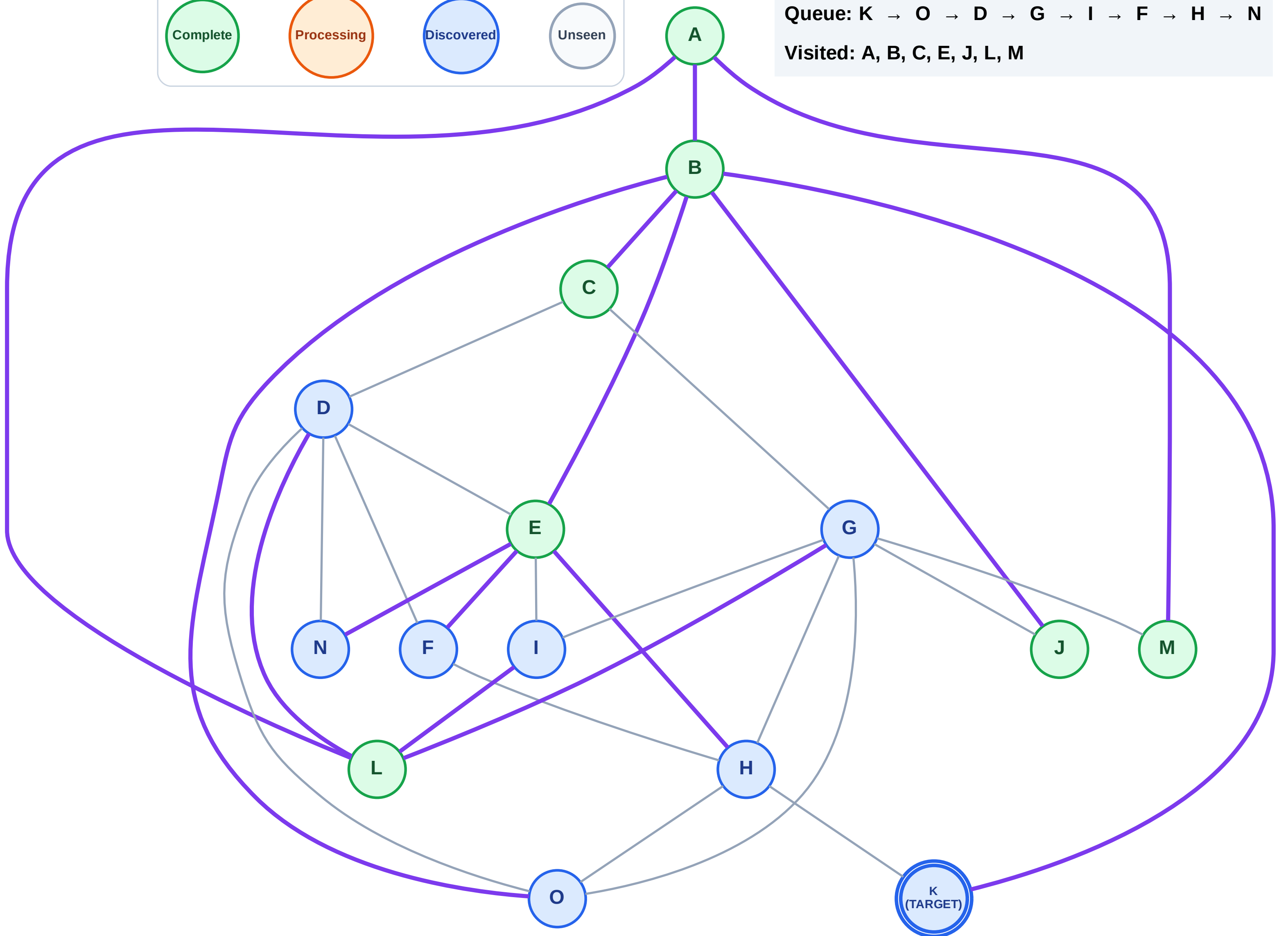
Step 15: No new neighbors from J

Legend



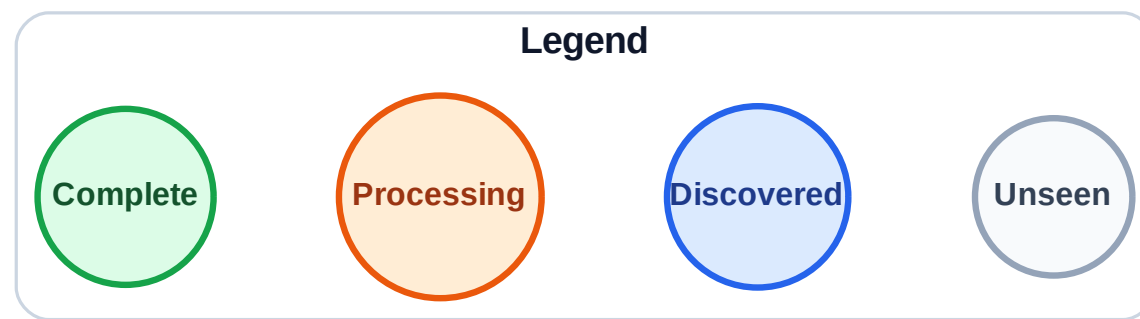
Queue: K → O → D → G → I → F → H → N

Visited: A, B, C, E, J, L, M



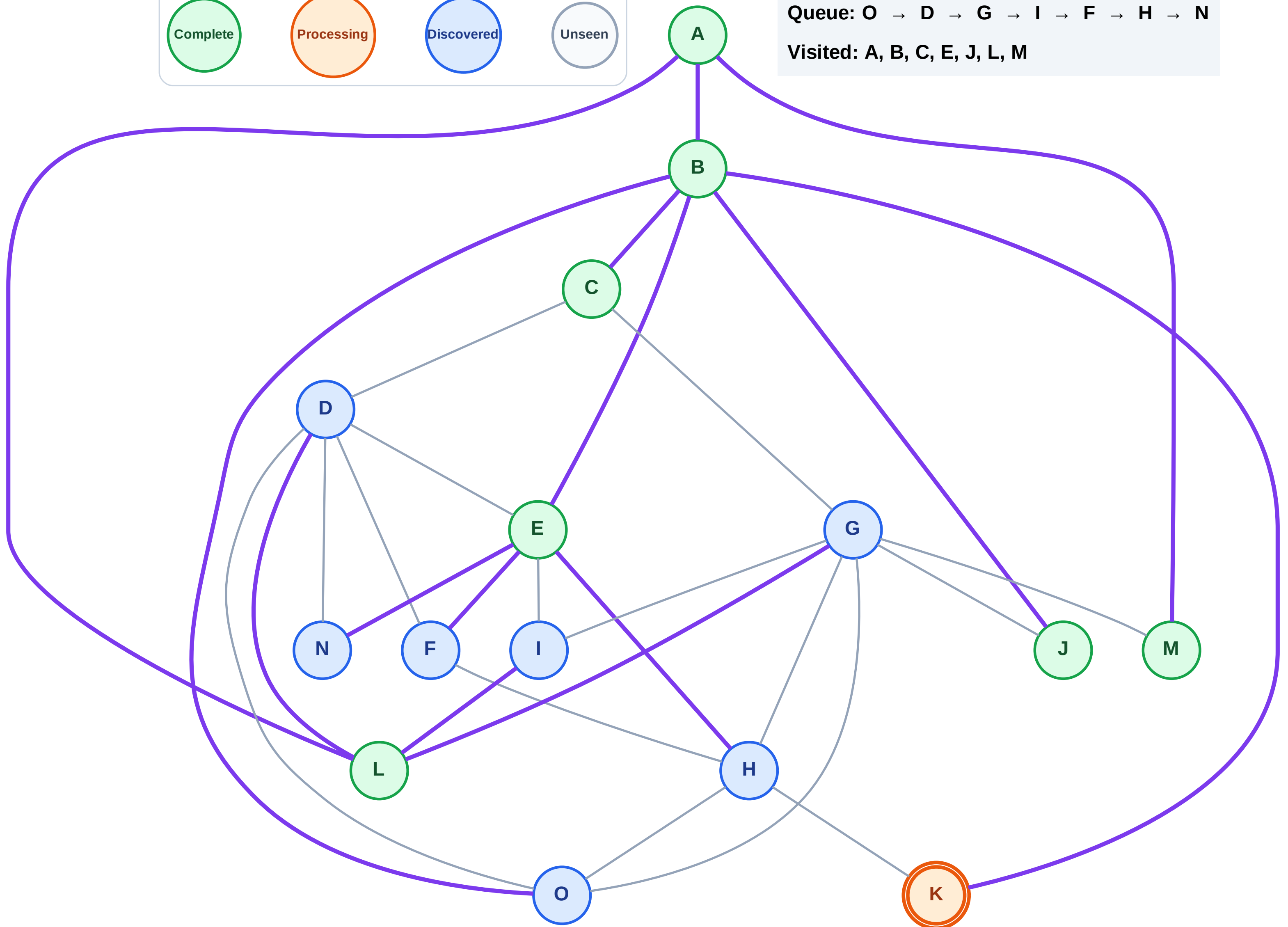
BFS Traversal

Step 16: Process node K



Queue: O → D → G → I → F → H → N

Visited: A, B, C, E, J, L, M



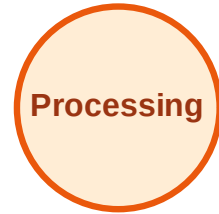
BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



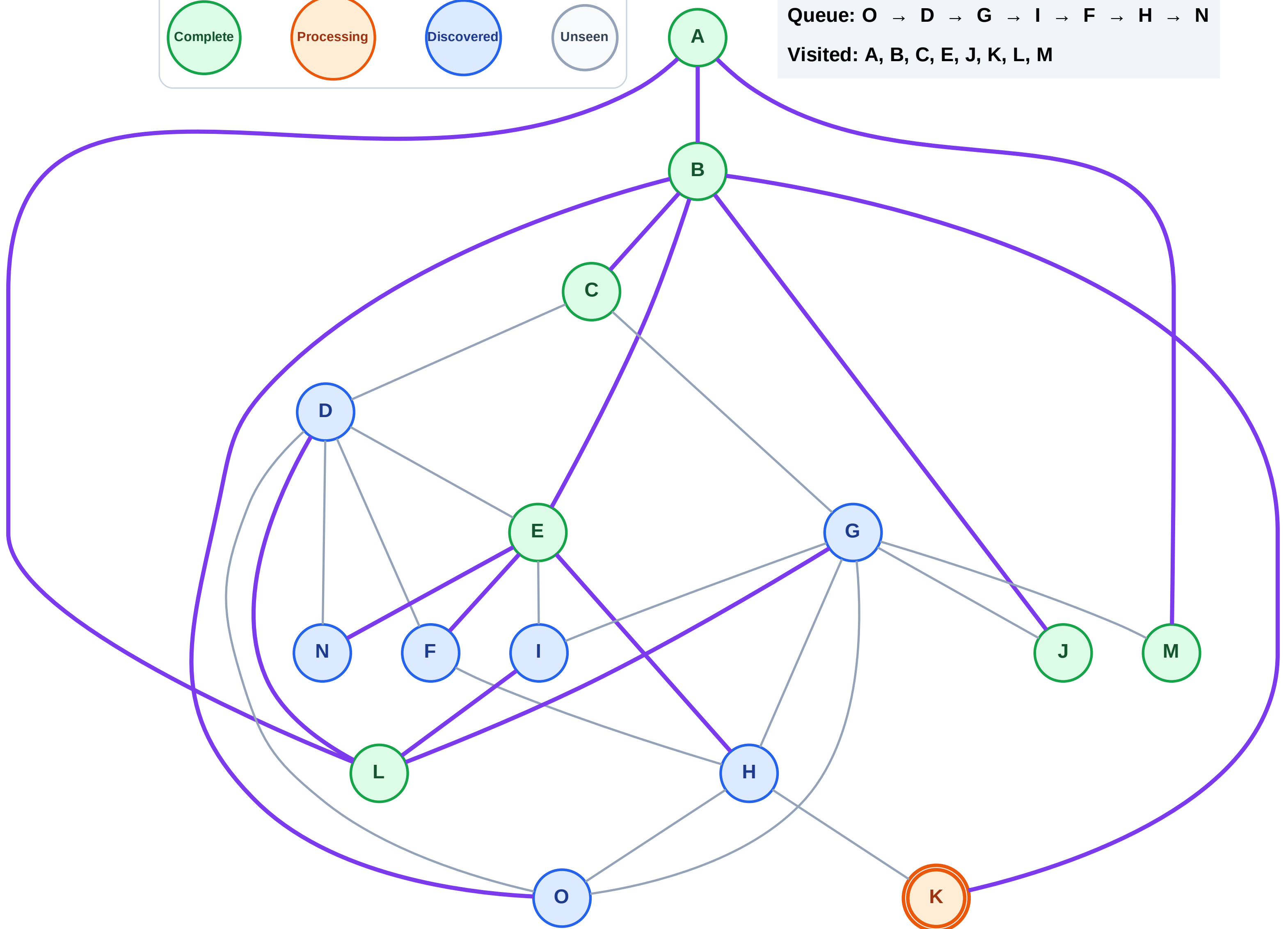
Discovered



Unseen

Queue: O → D → G → I → F → H → N

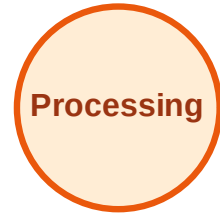
Visited: A, B, C, E, J, K, L, M



DFS Traversal

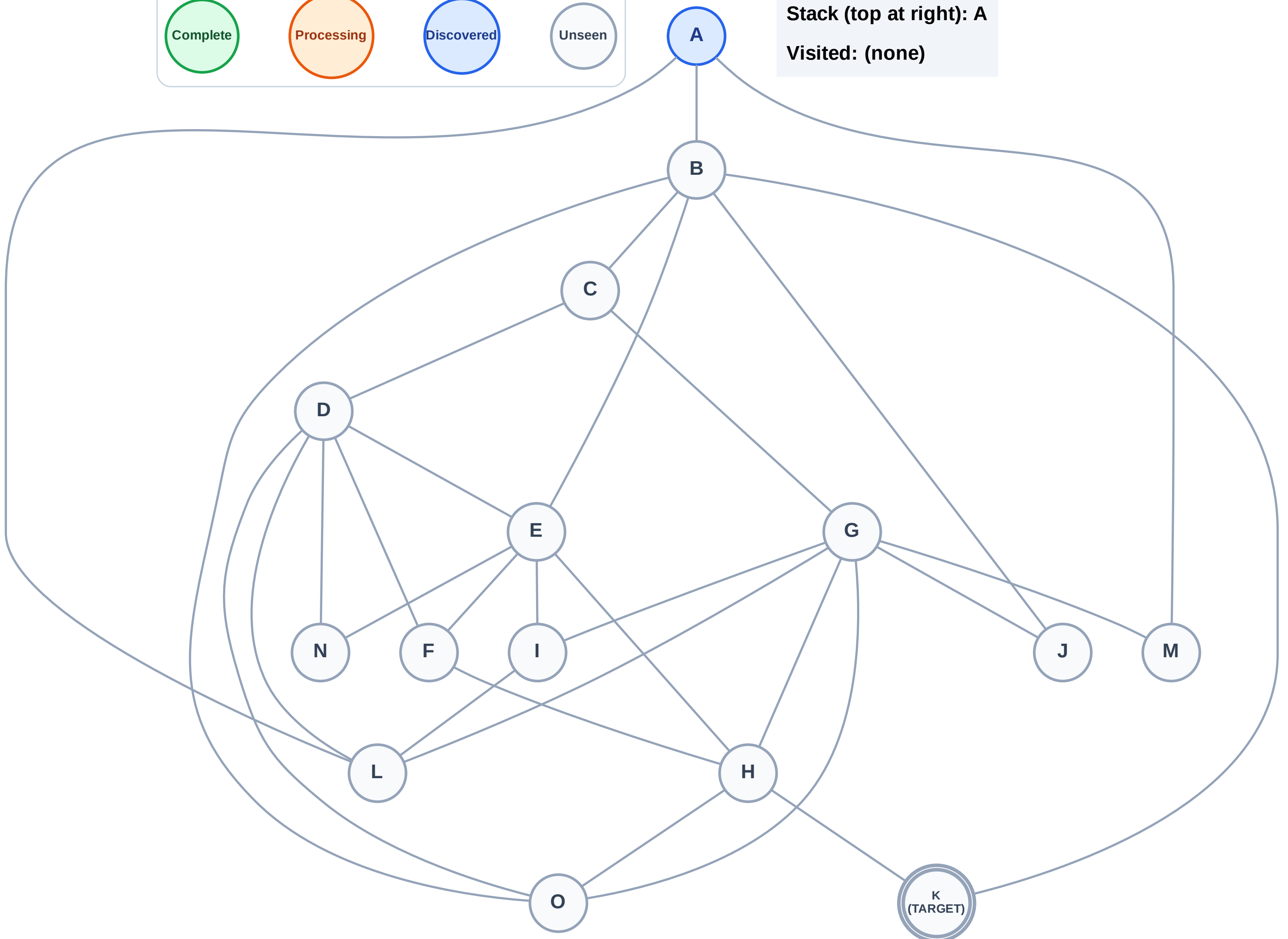
Step 1: Start at A (Seeking target node K)

Legend



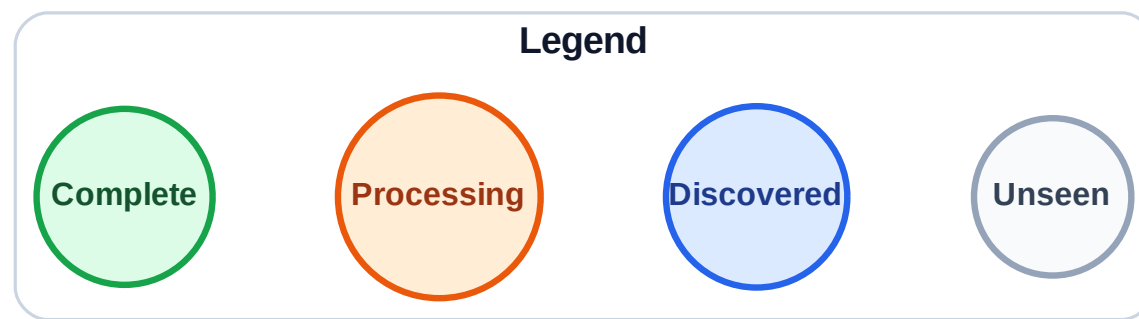
Stack (top at right): A

Visited: (none)



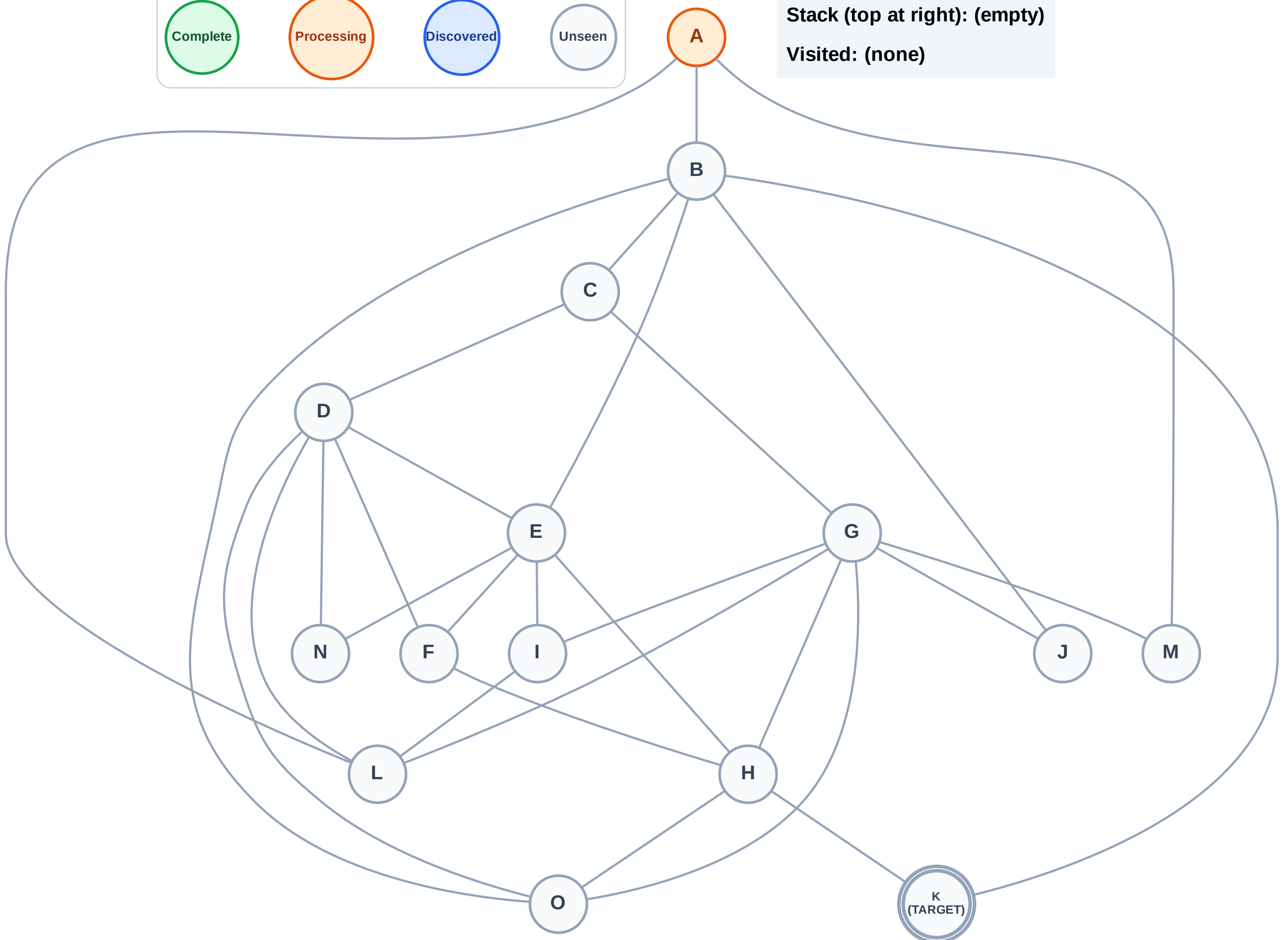
Step 2: Process node A

Step 2: Process node A



Stack (top at right): (empty)

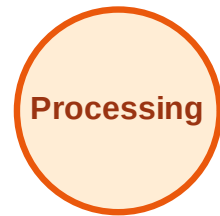
Visited: (none)



DFS Traversal

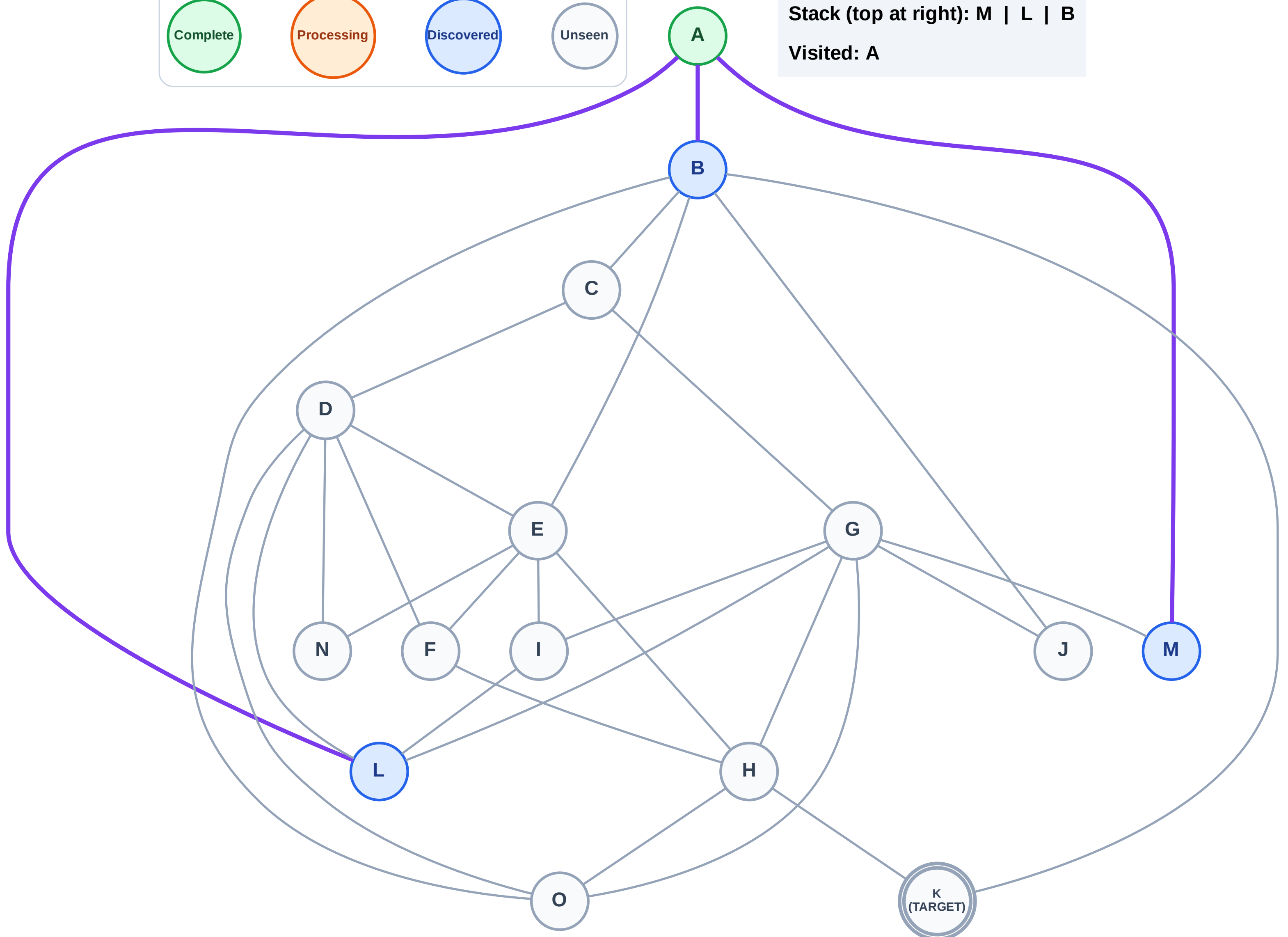
Step 3: Discover M, L, B from A

Legend



Stack (top at right): M | L | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

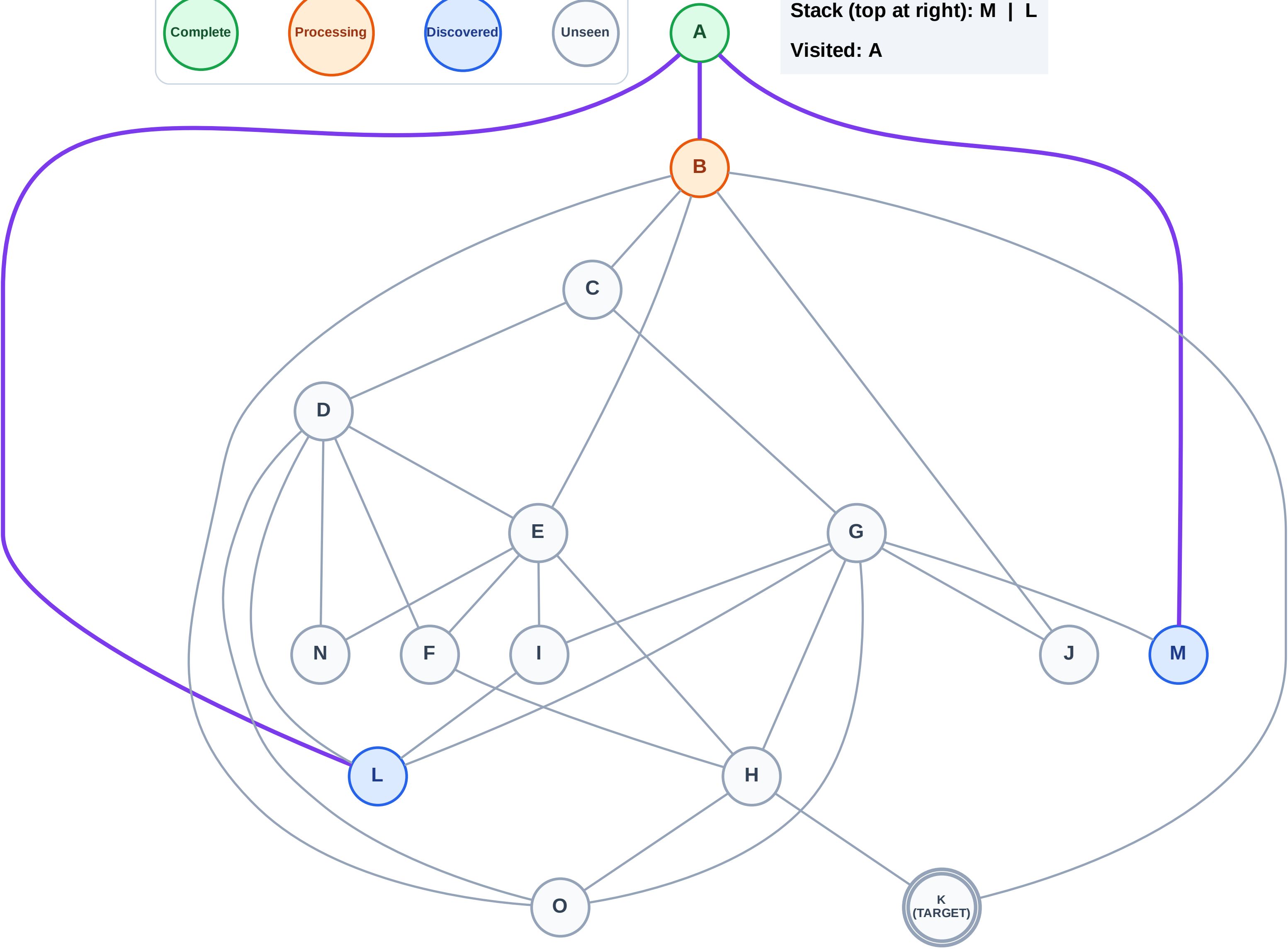
Processing

Discovered

Unseen

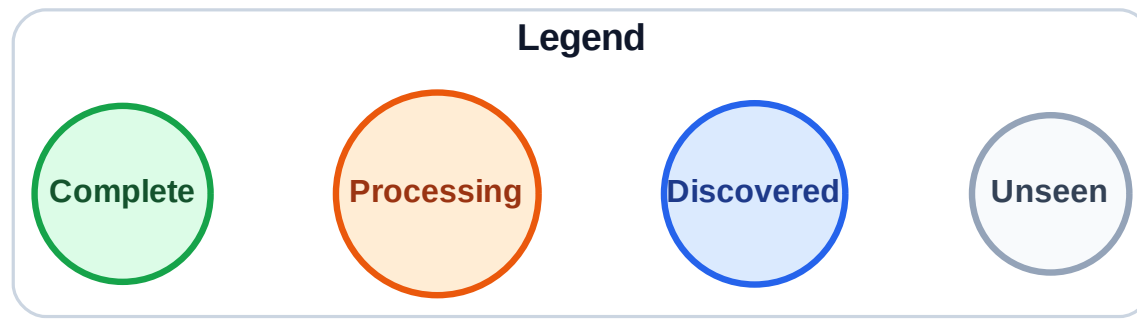
Stack (top at right): M | L

Visited: A

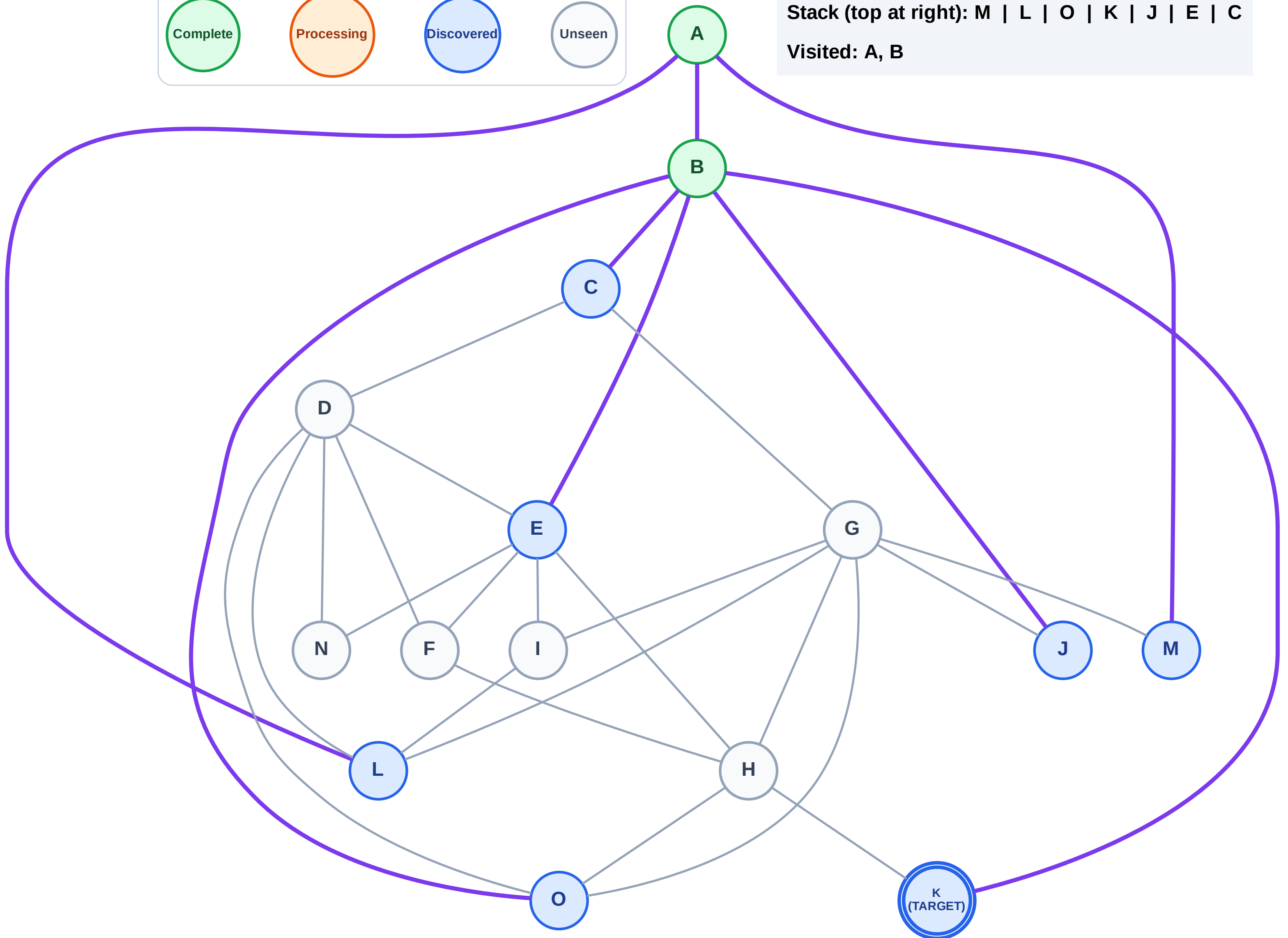


DFS Traversal

Step 5: Discover O, K, J, E, C from B

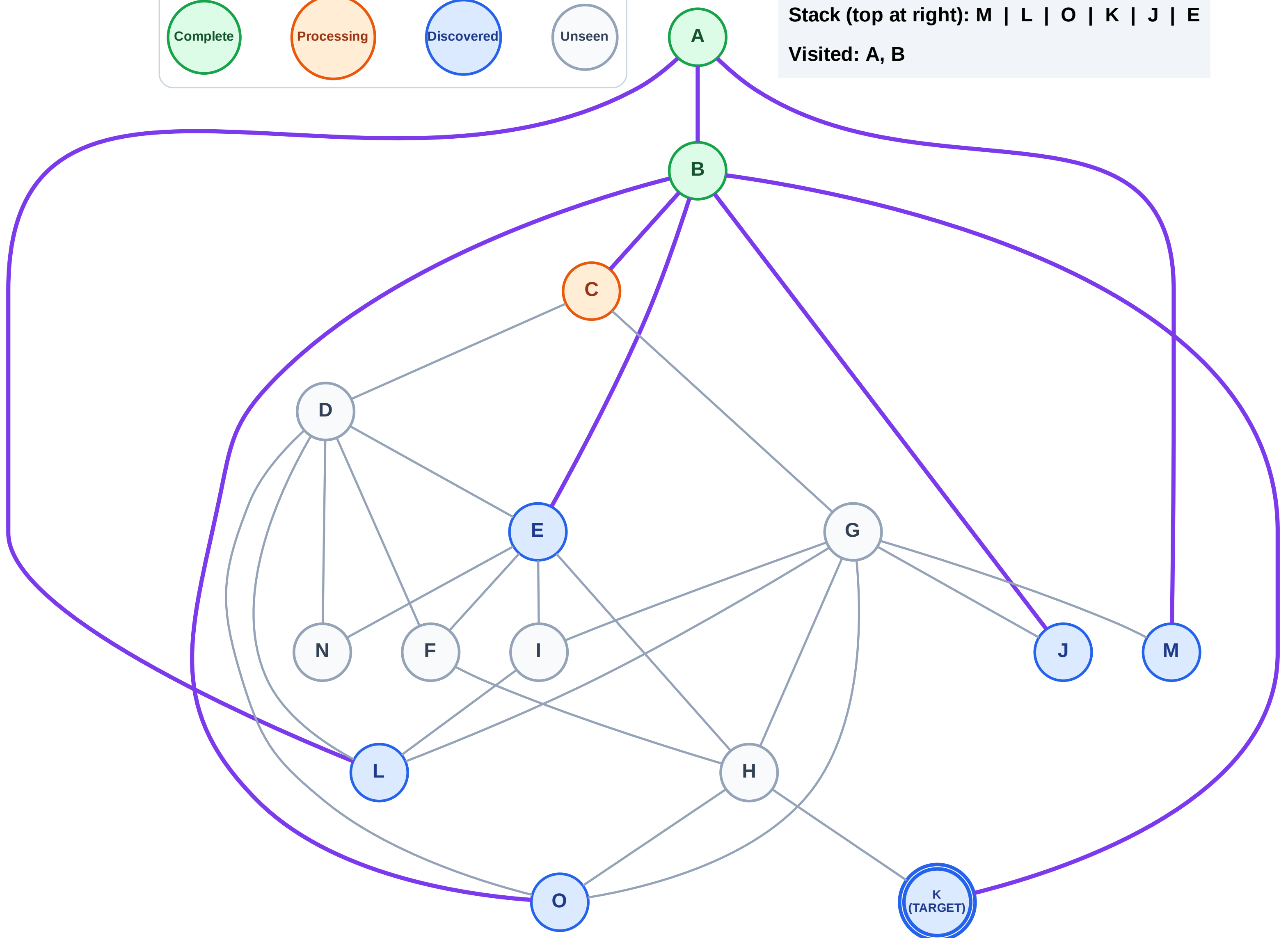
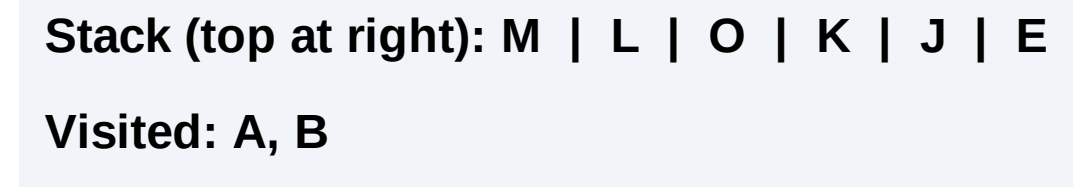


Stack (top at right): M | L | O | K | J | E | C
Visited: A, B



Step 6: Process node C

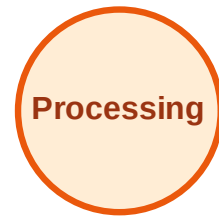
Step 6: Process node C



DFS Traversal

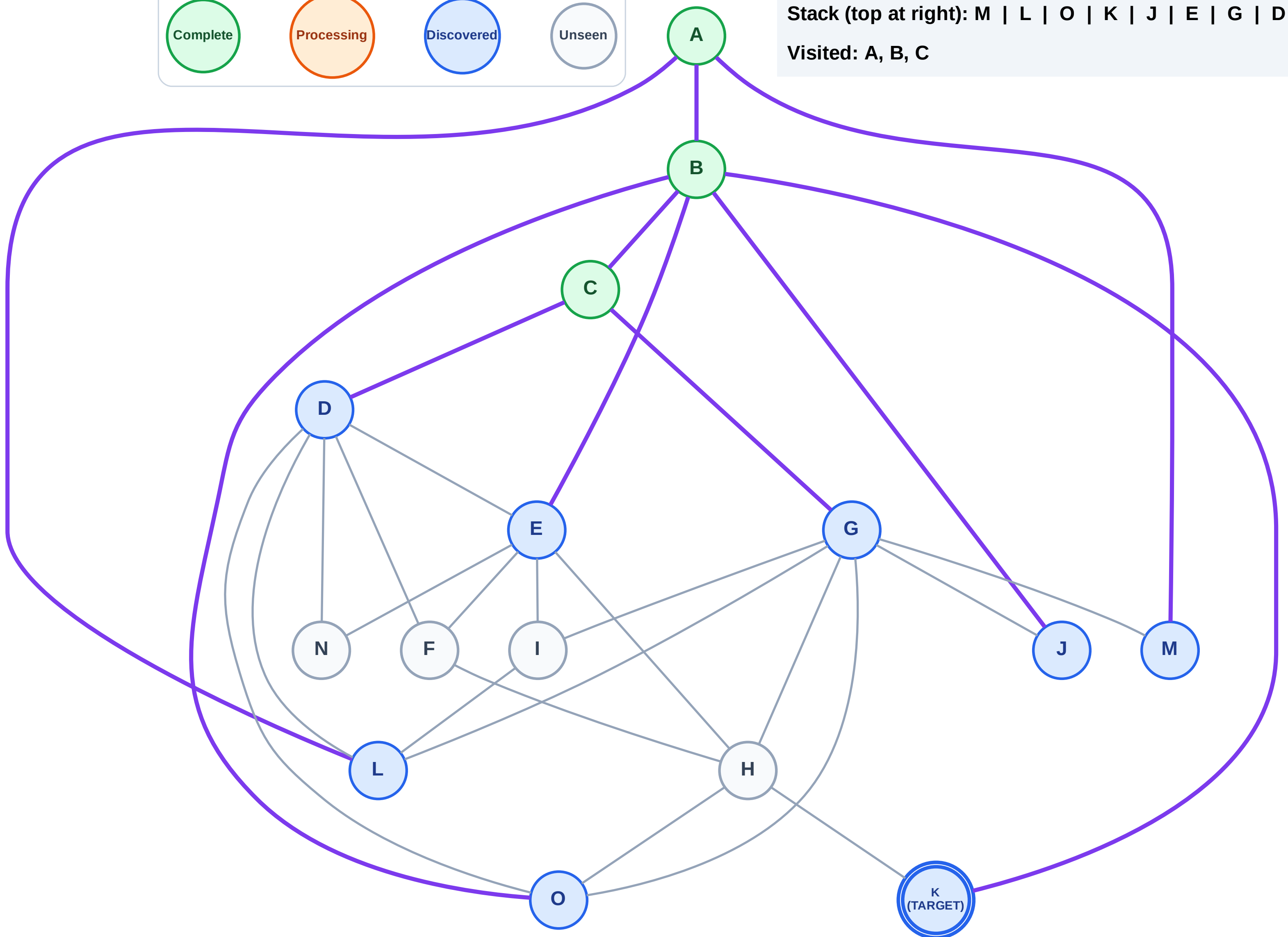
Step 7: Discover G, D from C

Legend



Stack (top at right): M | L | O | K | J | E | G | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

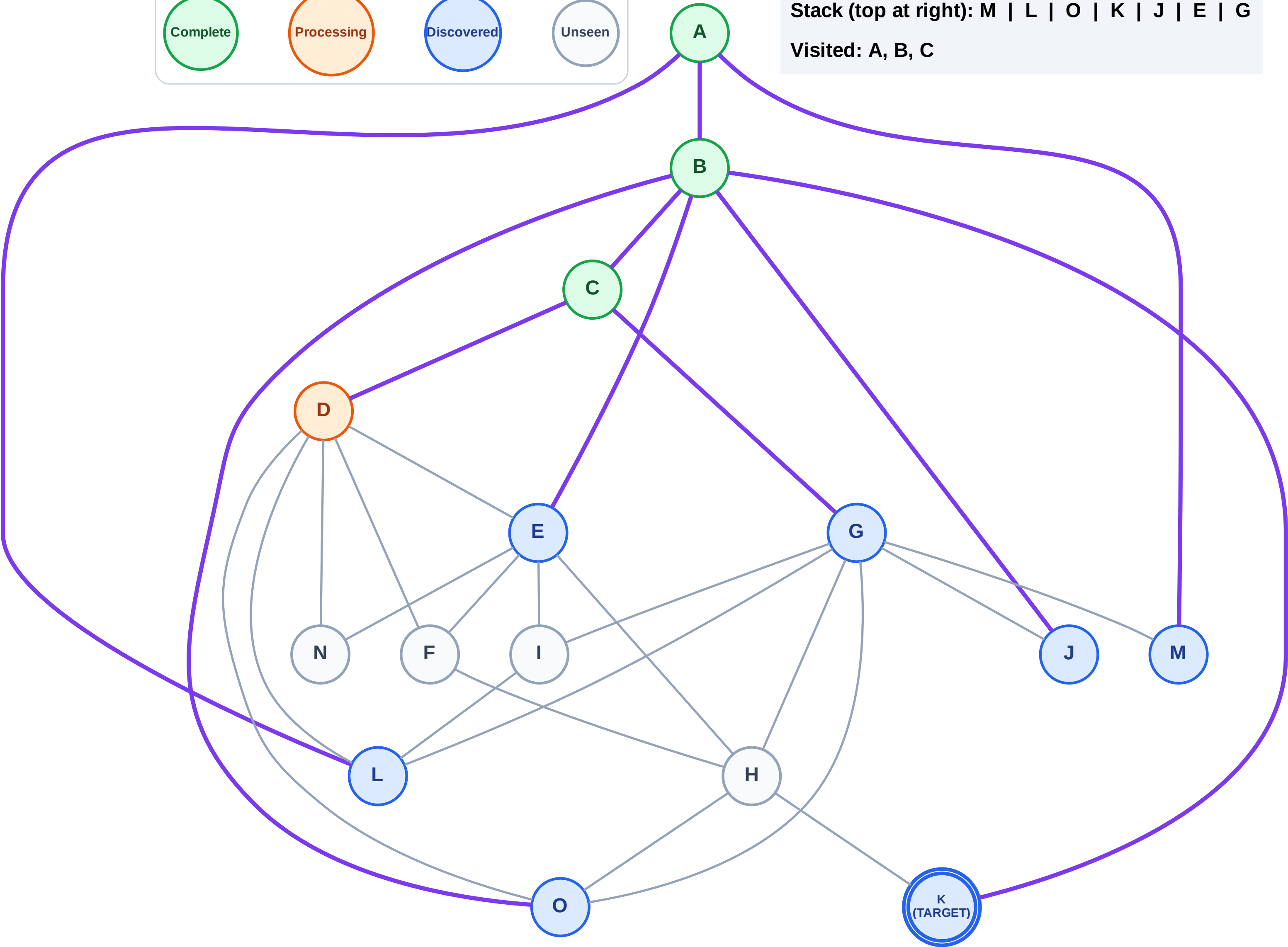
Processing

Discovered

Unseen

Stack (top at right): M | L | O | K | J | E | G

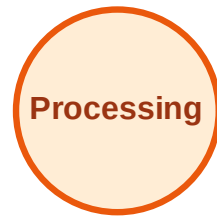
Visited: A, B, C



DFS Traversal

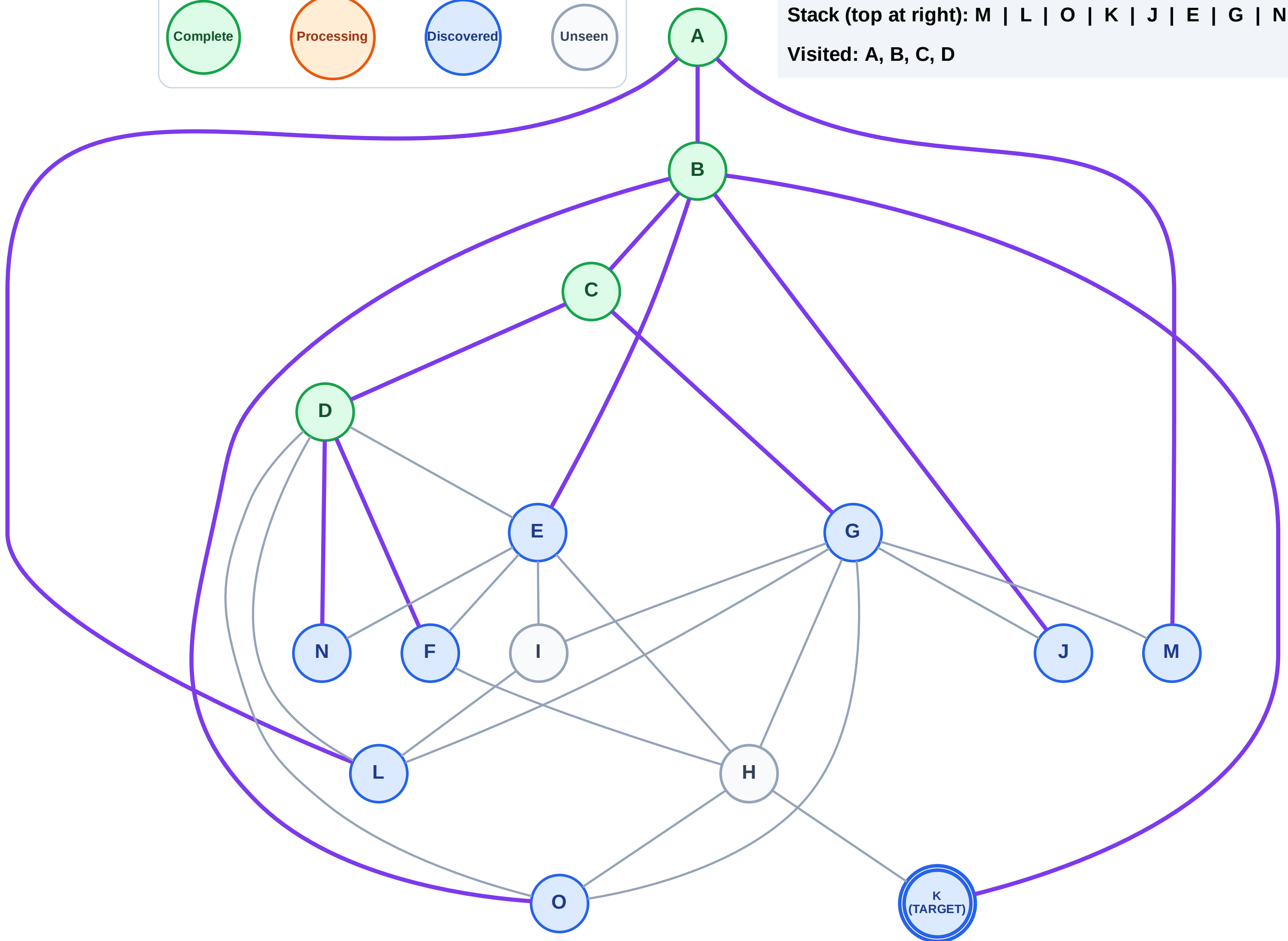
Step 9: Discover N, F from D

Legend



Stack (top at right): M | L | O | K | J | E | G | N | F

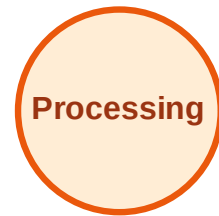
Visited: A, B, C, D



DFS Traversal

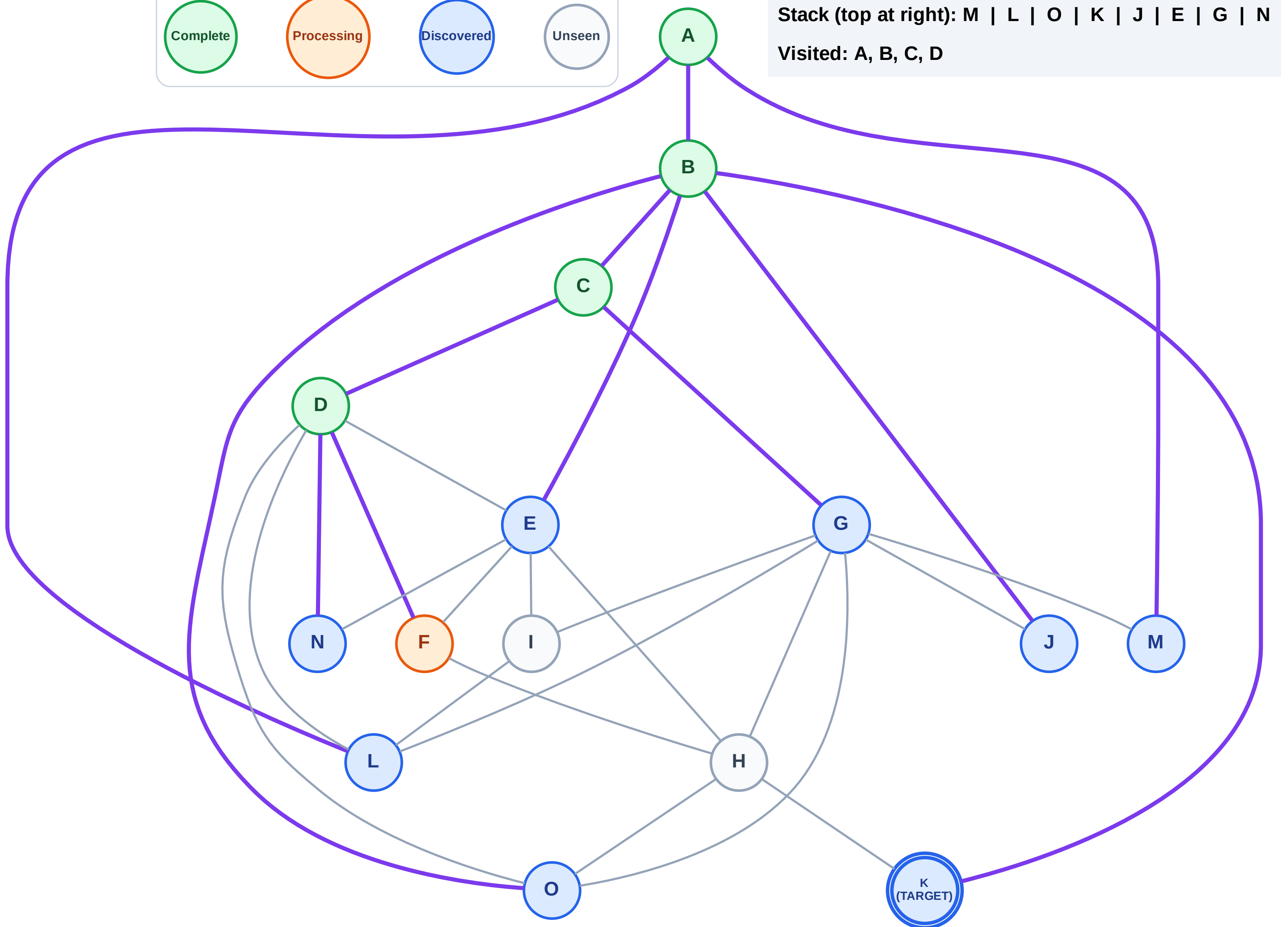
Step 10: Process node F

Legend

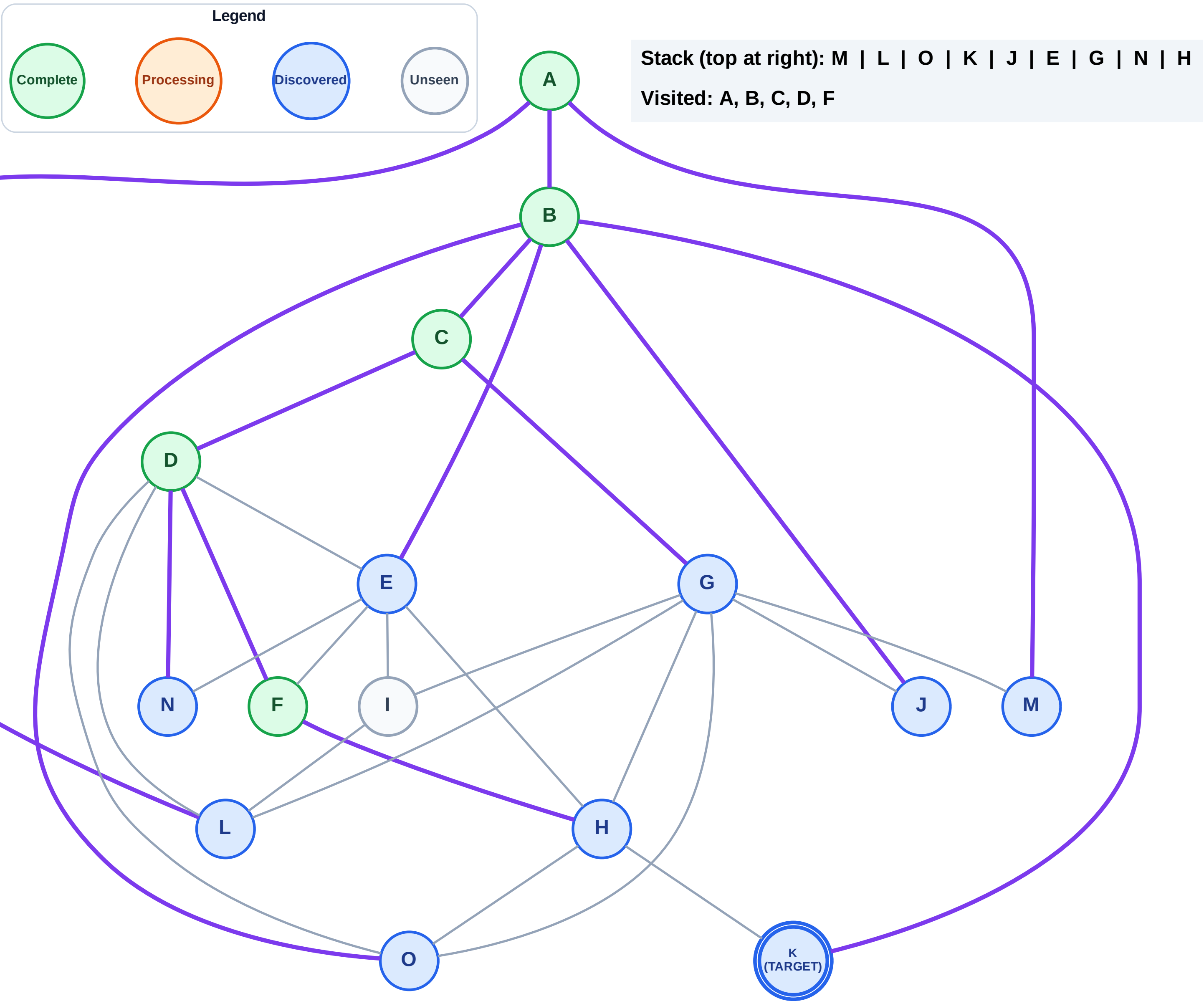


Stack (top at right): M | L | O | K | J | E | G | N

Visited: A, B, C, D

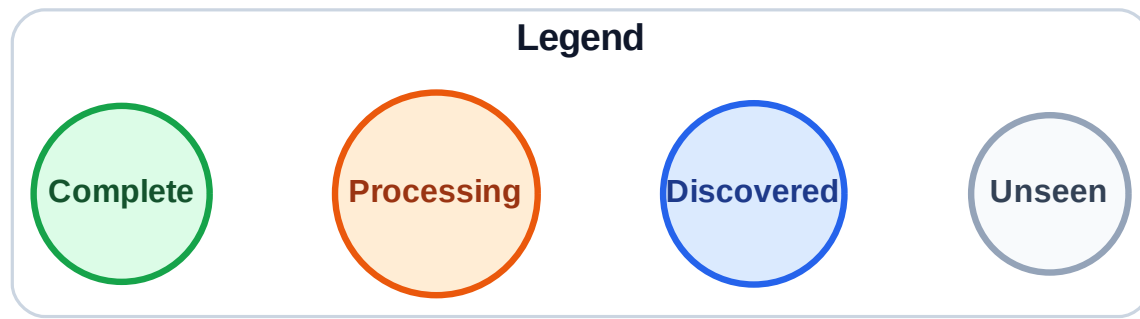


Step 11: Discover H from F

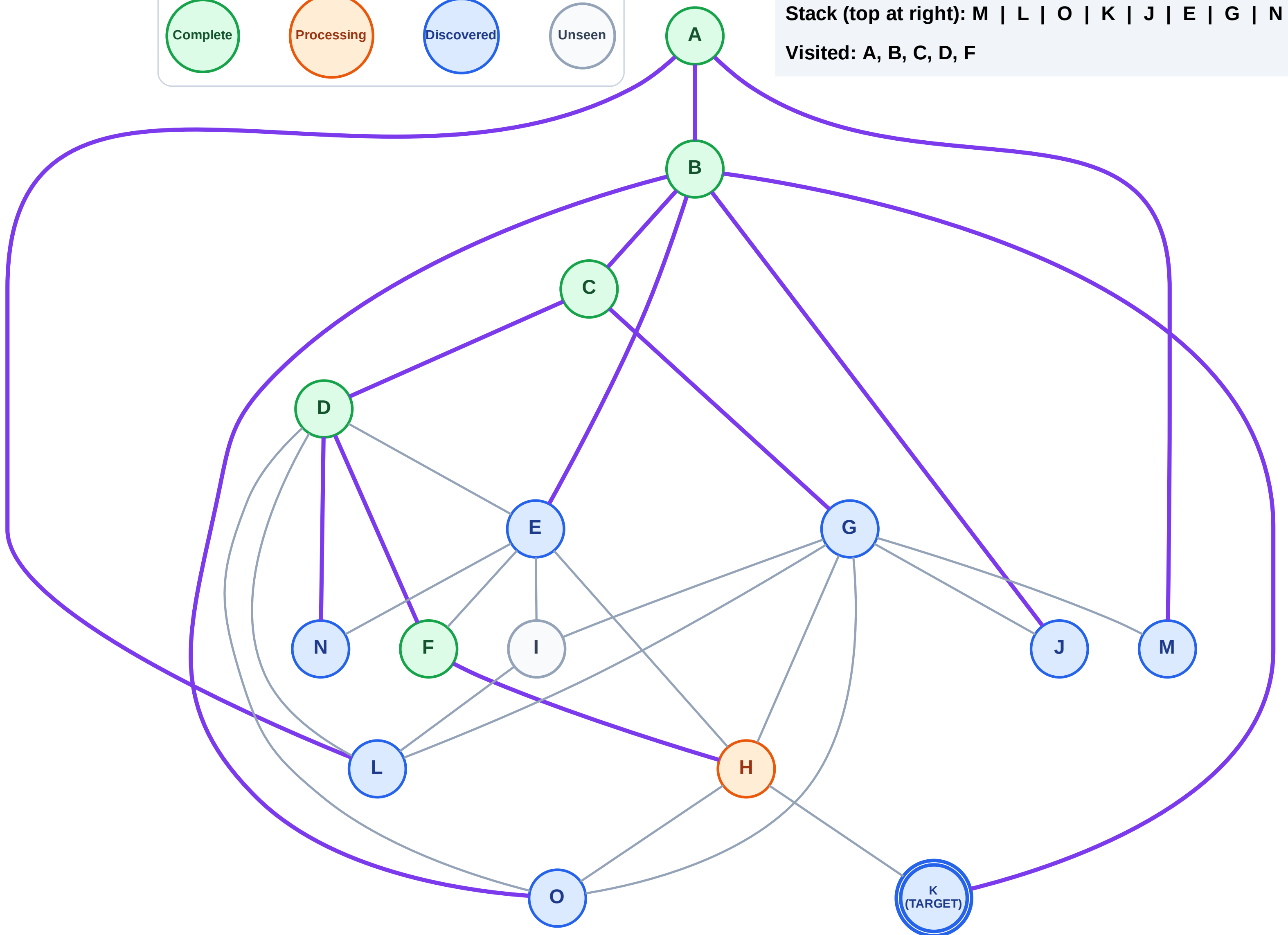


DFS Traversal

Step 12: Process node H



Stack (top at right): M | L | O | K | J | E | G | N
Visited: A, B, C, D, F



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

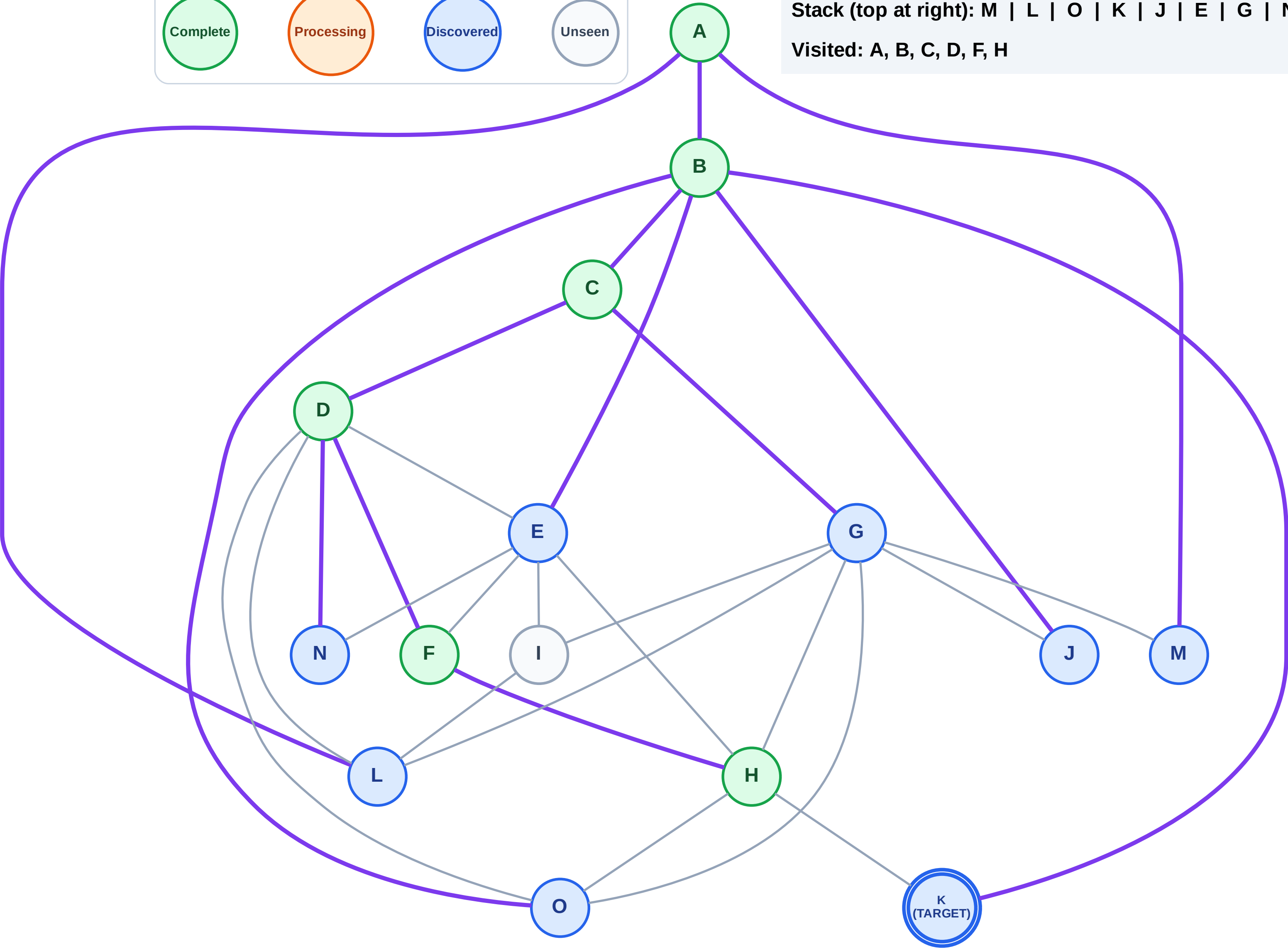
Processing

Discovered

Unseen

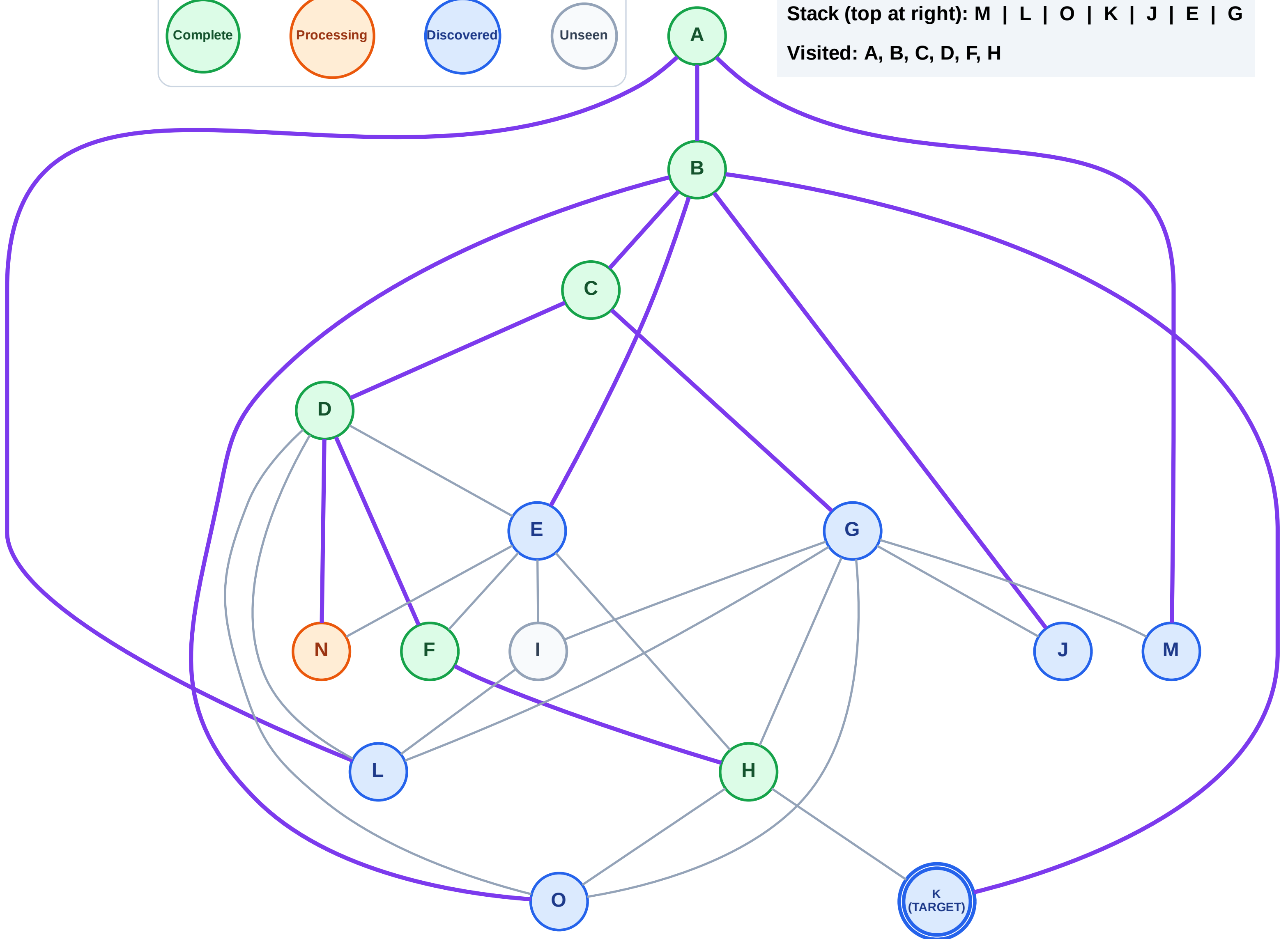
Stack (top at right): M | L | O | K | J | E | G | N

Visited: A, B, C, D, F, H



Step 14: Process node N

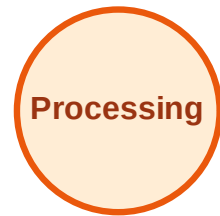
Step 14: Process node N



DFS Traversal

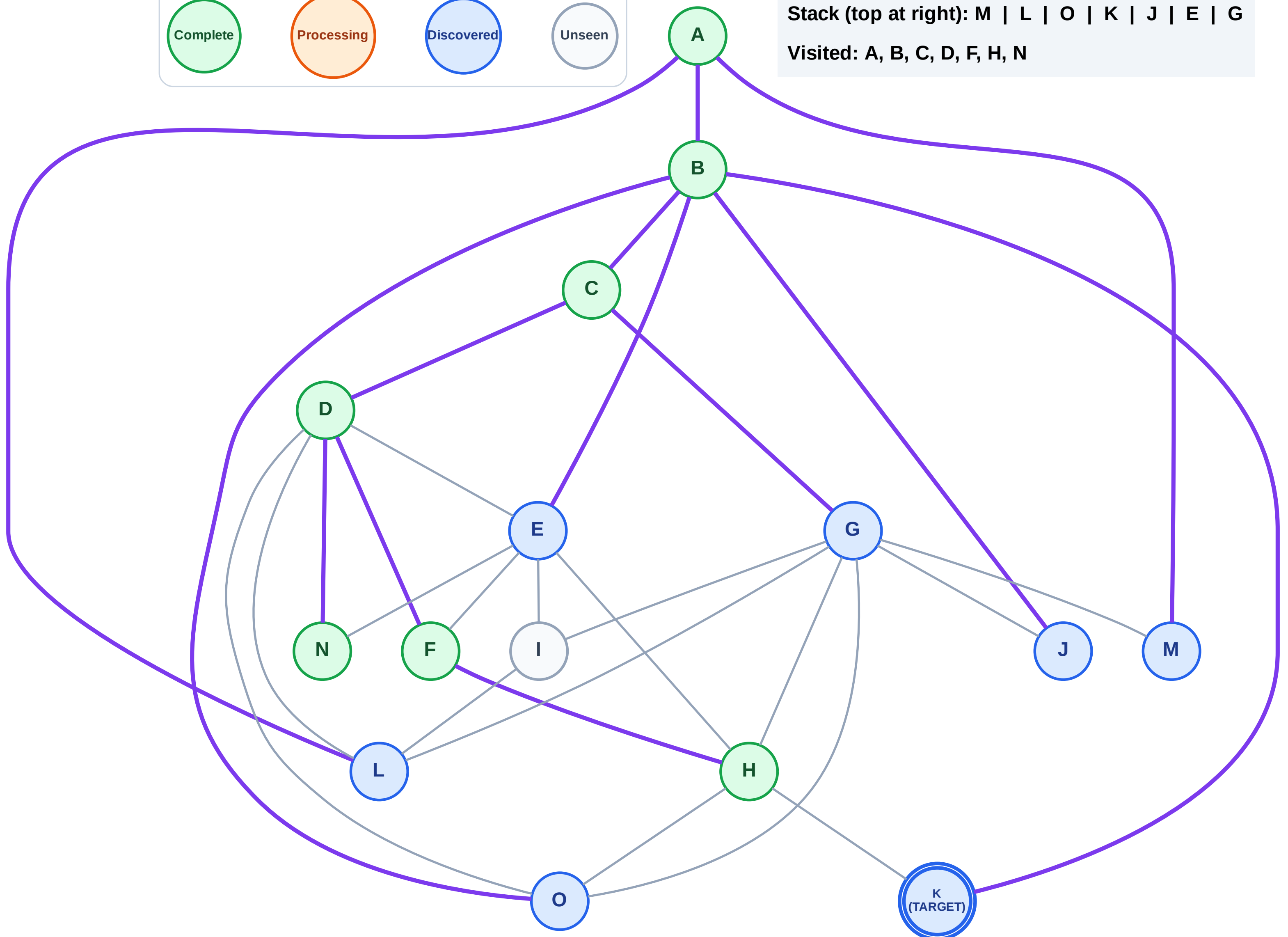
Step 15: No new neighbors from N

Legend



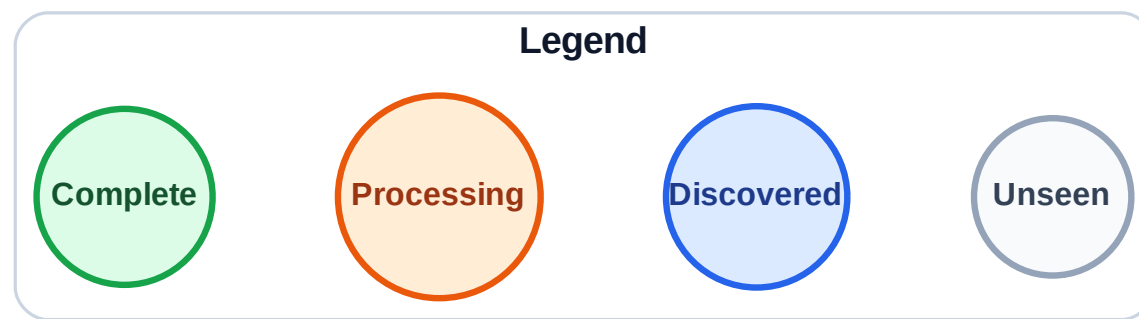
Stack (top at right): M | L | O | K | J | E | G

Visited: A, B, C, D, F, H, N



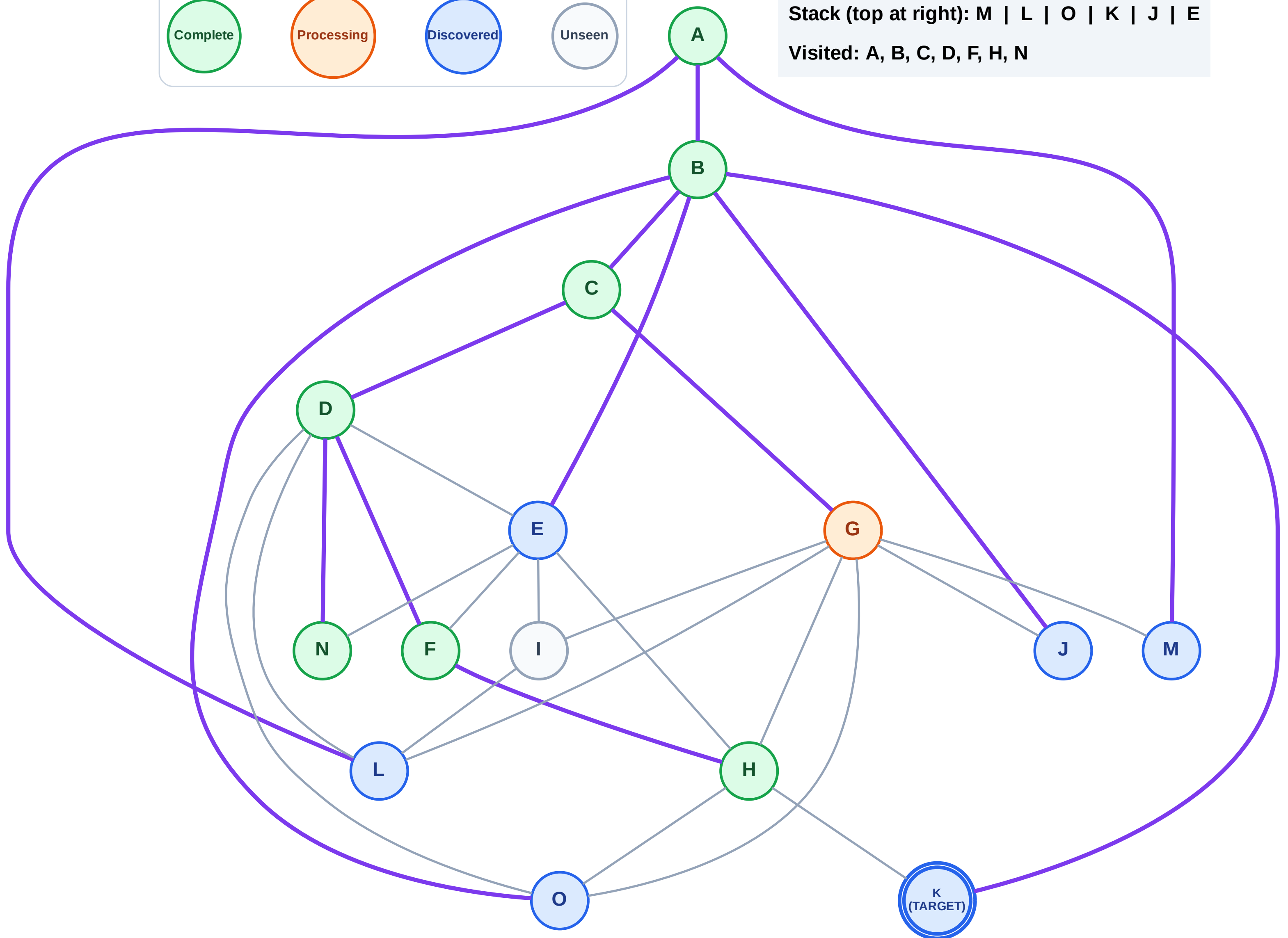
DFS Traversal

Step 16: Process node G



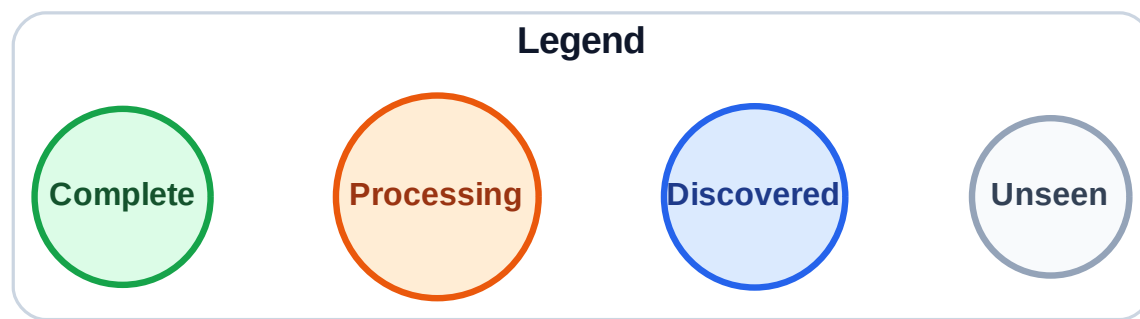
Stack (top at right): M | L | O | K | J | E

Visited: A, B, C, D, F, H, N



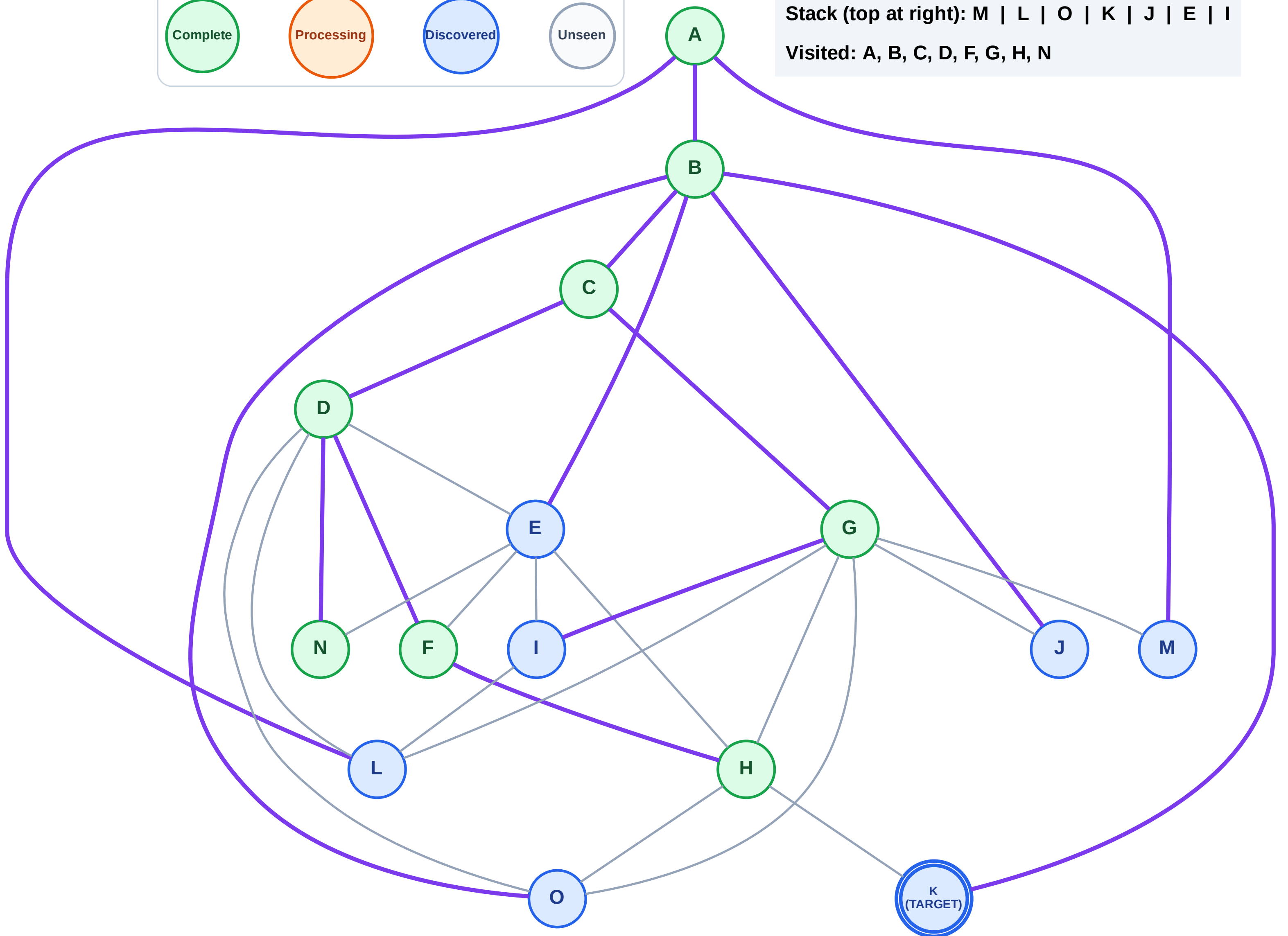
DFS Traversal

Step 17: Discover I from G



Stack (top at right): M | L | O | K | J | E | I

Visited: A, B, C, D, F, G, H, N



DFS Traversal

Step 18: Process node I

Legend

Complete

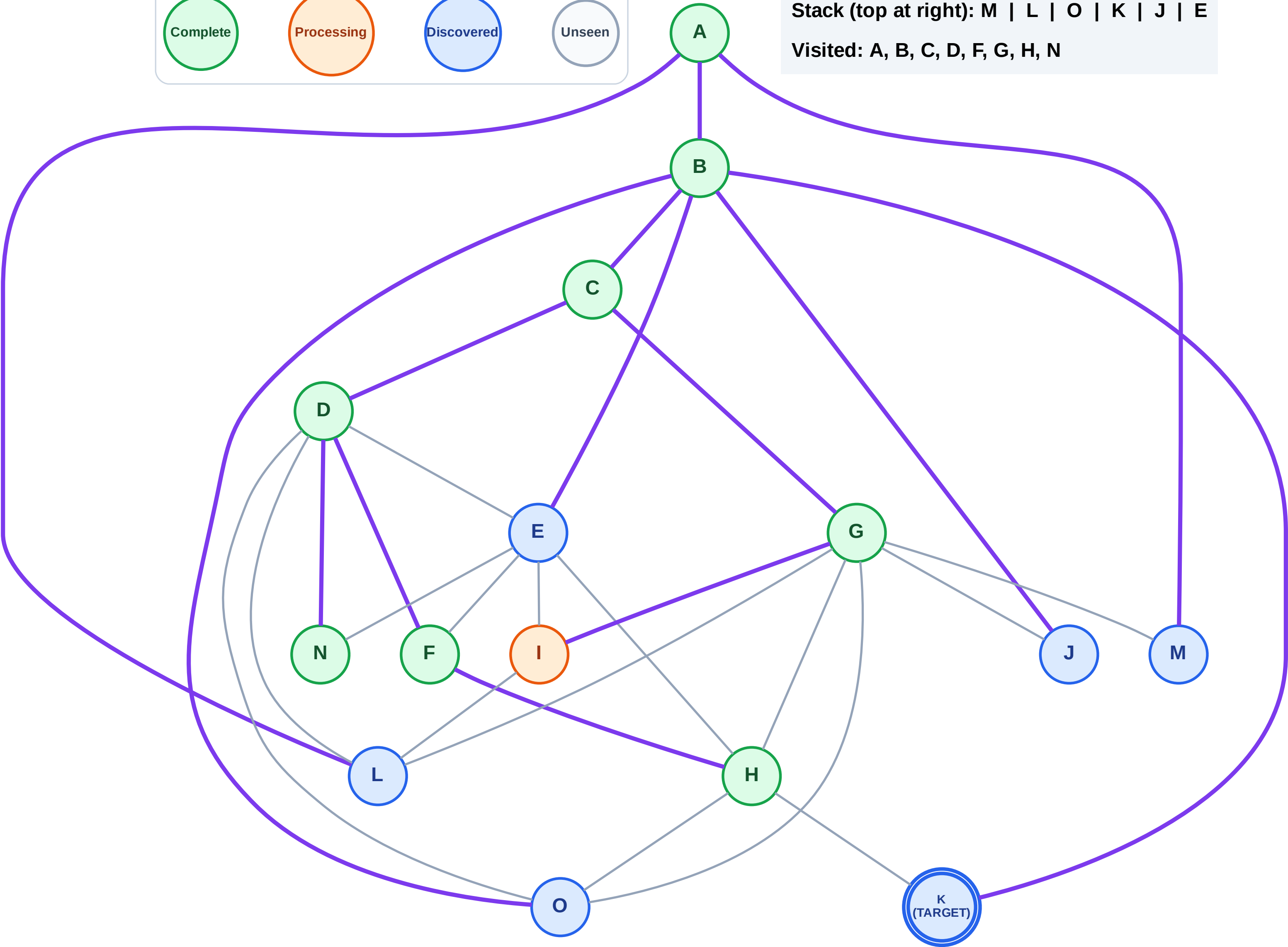
Processing

Discovered

Unseen

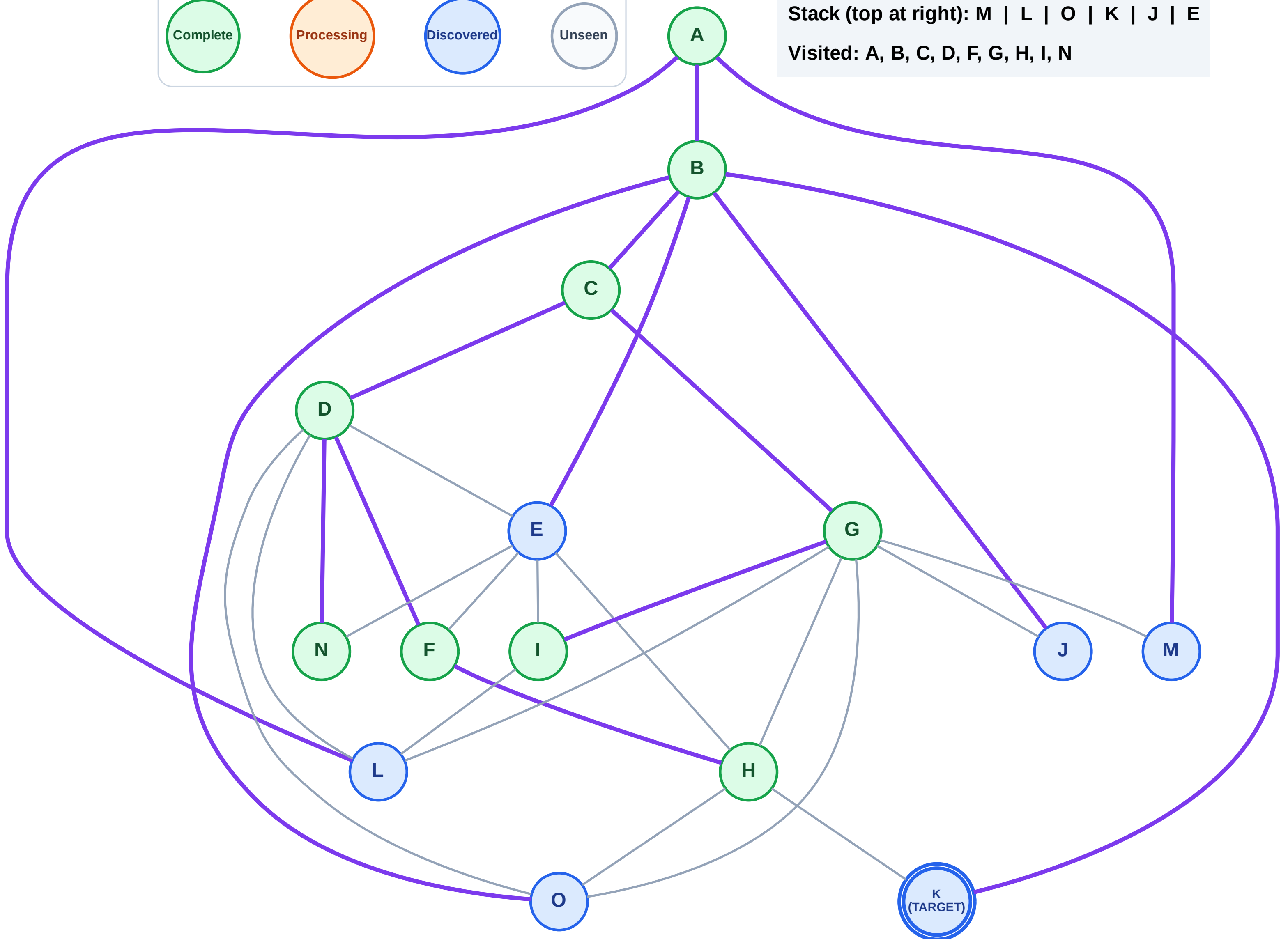
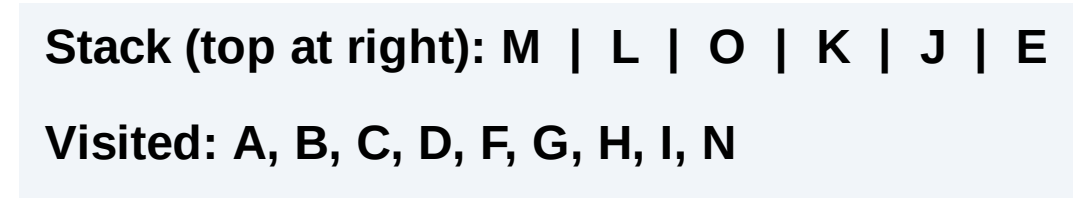
Stack (top at right): M | L | O | K | J | E

Visited: A, B, C, D, F, G, H, N



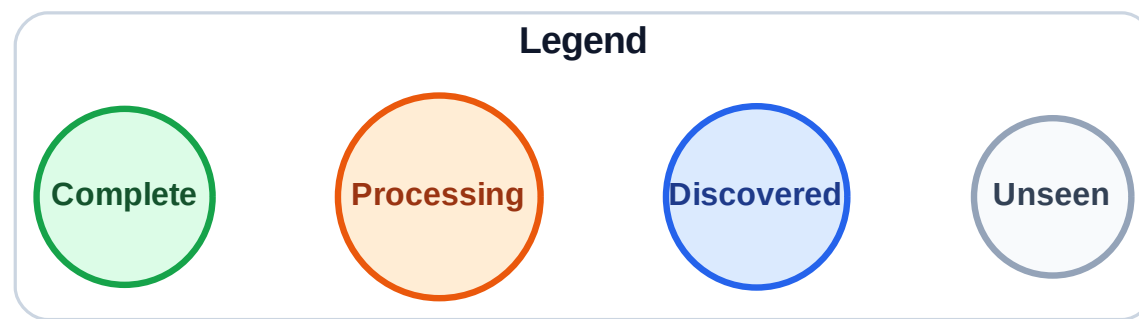
Step 19: No new neighbors from I

Step 19: No new neighbors from I

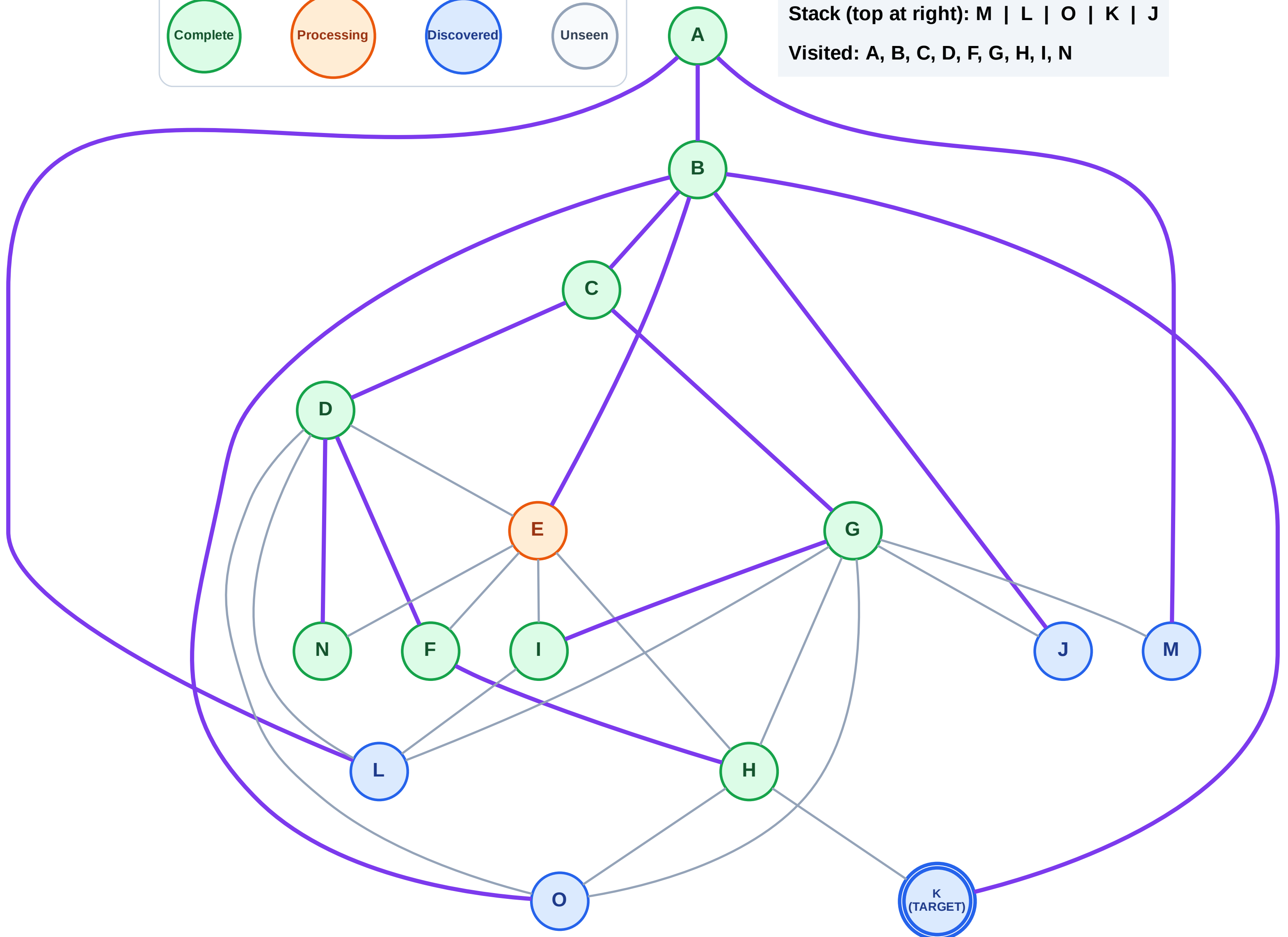


DFS Traversal

Step 20: Process node E

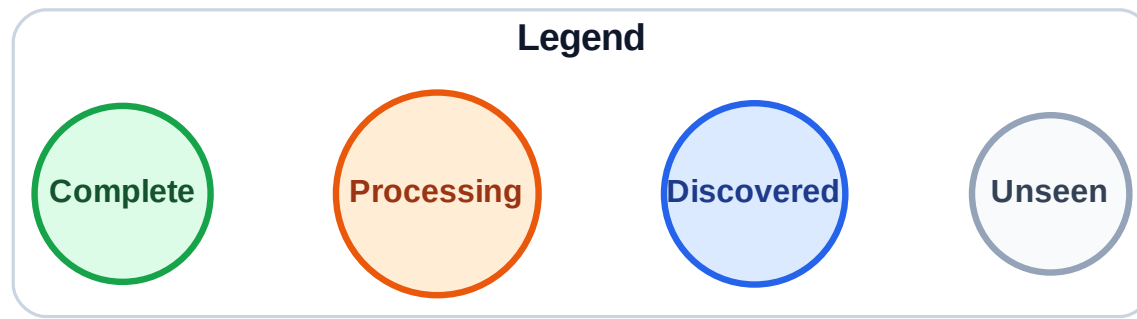


Stack (top at right): M | L | O | K | J
Visited: A, B, C, D, F, G, H, I, N

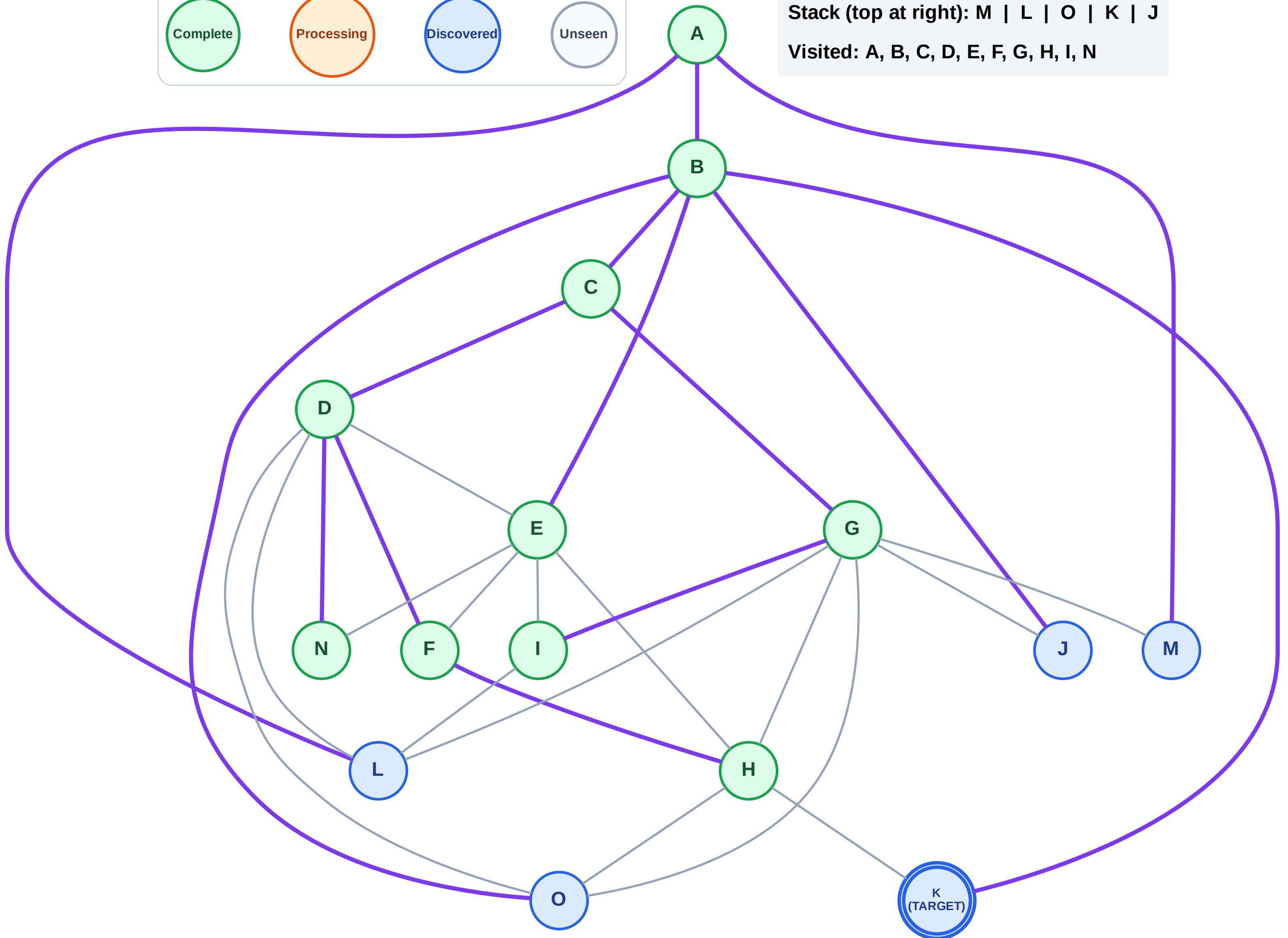


DFS Traversal

Step 21: No new neighbors from E



Stack (top at right): M | L | O | K | J
Visited: A, B, C, D, E, F, G, H, I, N



DFS Traversal

Step 22: Process node J

Legend

Complete

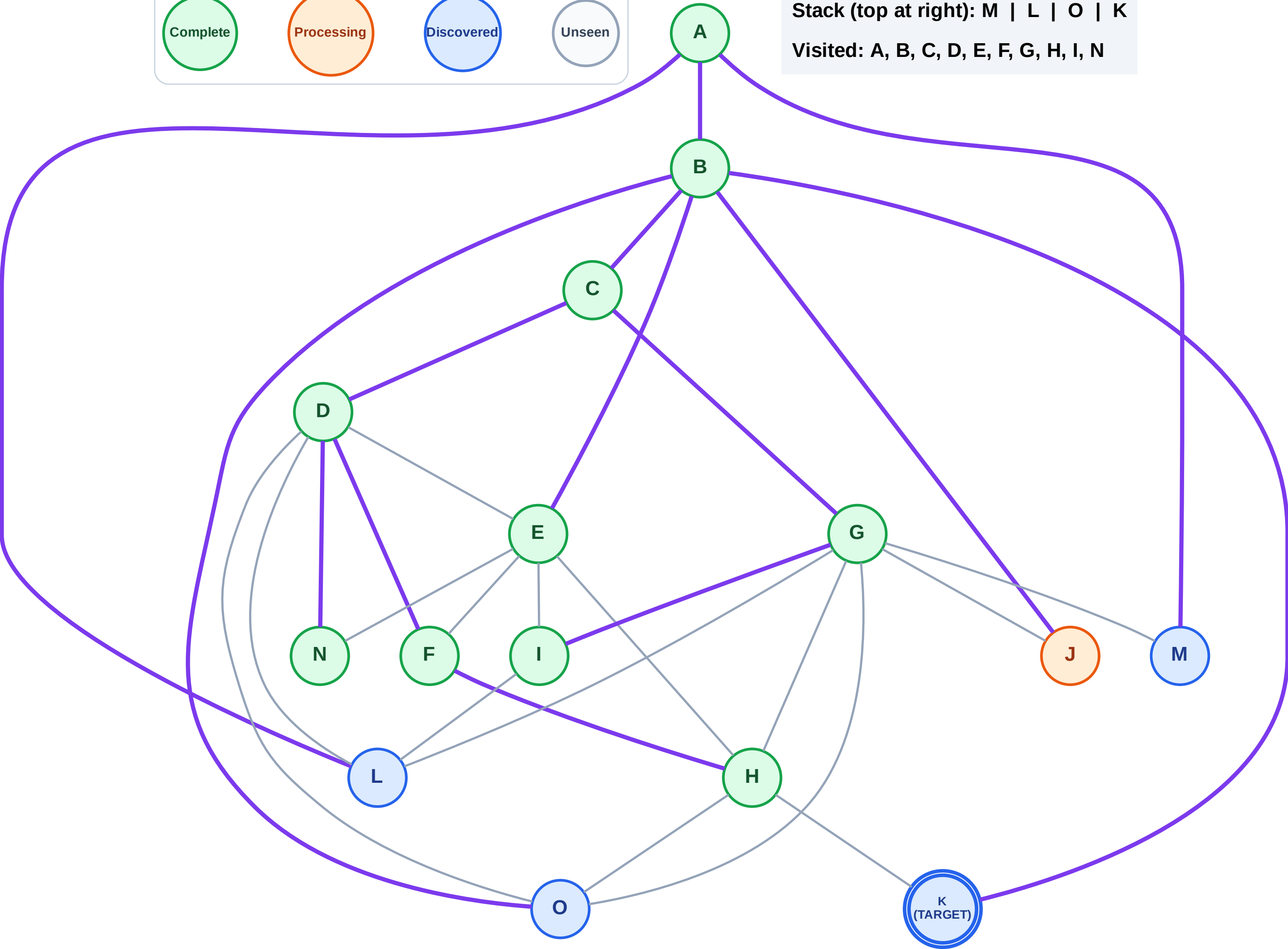
Processing

Discovered

Unseen

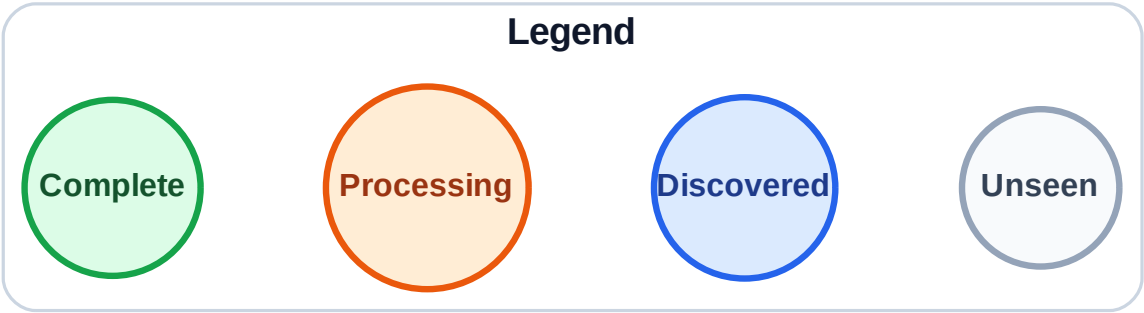
Stack (top at right): M | L | O | K

Visited: A, B, C, D, E, F, G, H, I, N

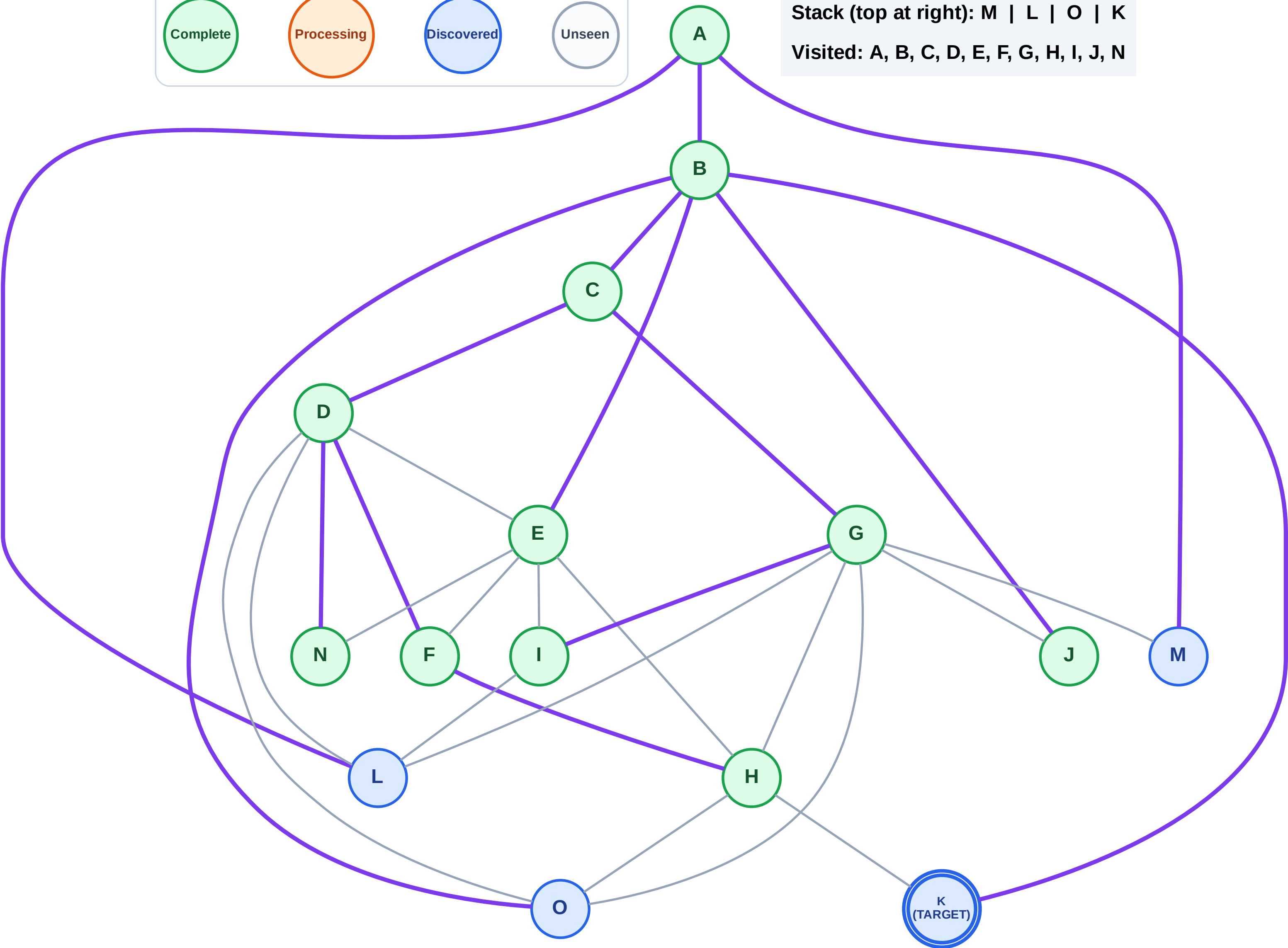


DFS Traversal

Step 23: No new neighbors from J

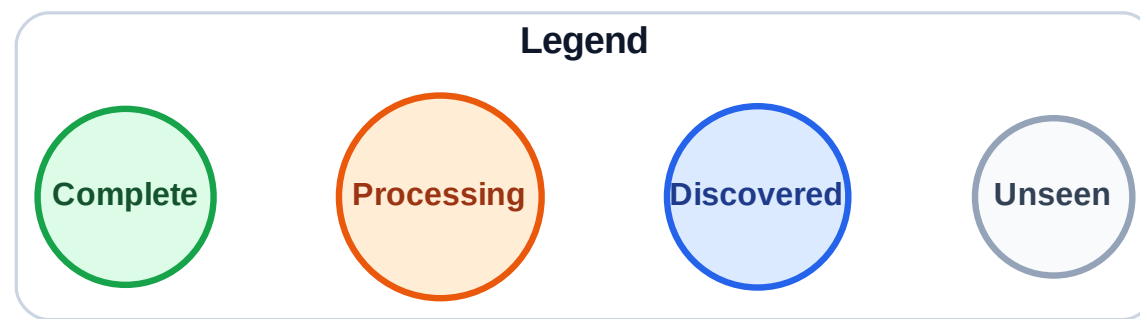


Stack (top at right): M | L | O | K
Visited: A, B, C, D, E, F, G, H, I, J, N

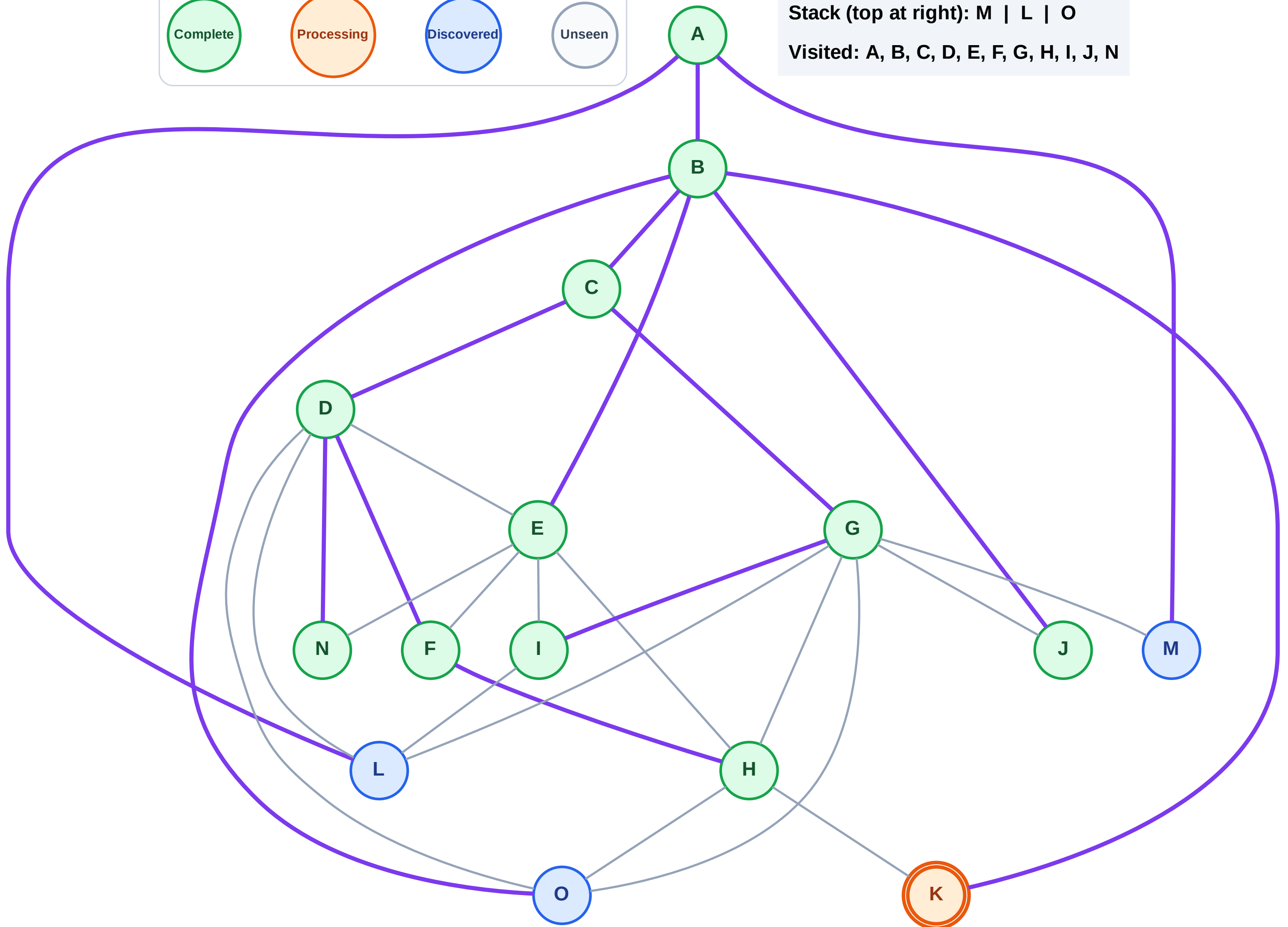


DFS Traversal

Step 24: Process node K



Stack (top at right): M | L | O
Visited: A, B, C, D, E, F, G, H, I, J, N



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | O
Visited: A, B, C, D, E, F, G, H, I, J, K, N

